Advanced Representation Learning
(고급표현학습)

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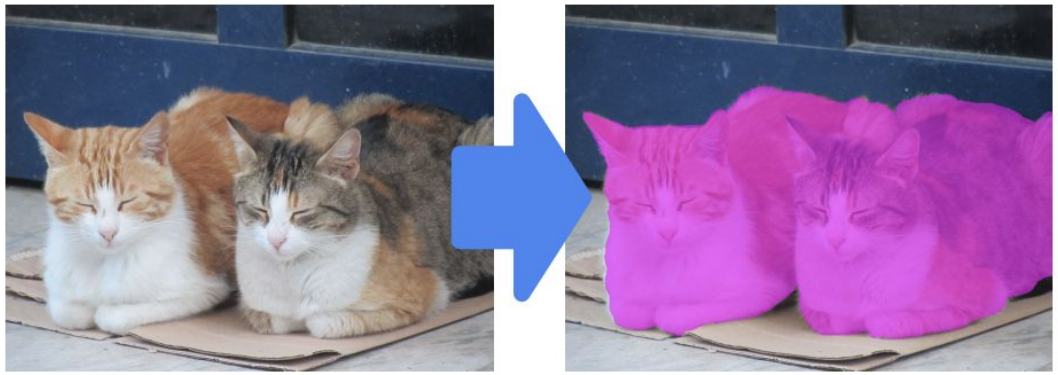
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- 2. From Image-Level Labels to Pixel-Level Labels
- 3. Webly Supervised Learning
- 4. Boxly Supervised Learning

Weakly-Supervised Semantic Segmentation

Fully Supervised vs Weakly Supervised

Fully supervised → **Directly** supervised



Weakly supervised → **Indirectly** supervised



Slide credit: Rodrigo Benenson

There are only two sources of information

- **Priors**: what do I believe to be true independent of any particular image sample
- **Hints**: indirect supervision received for each image

Slide credit: Rodrigo Benenson

Priors

- Size, Location, Shape





Slide credit: Rodrigo Benenson

Priors

- Number of instances, Contrast

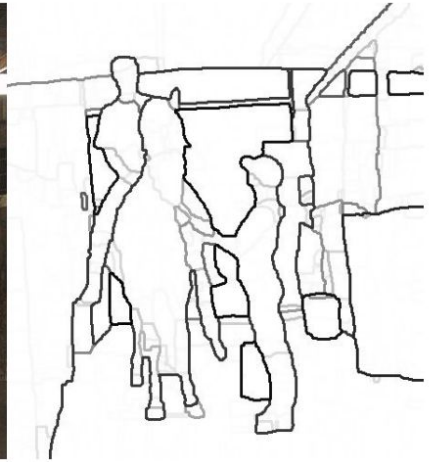


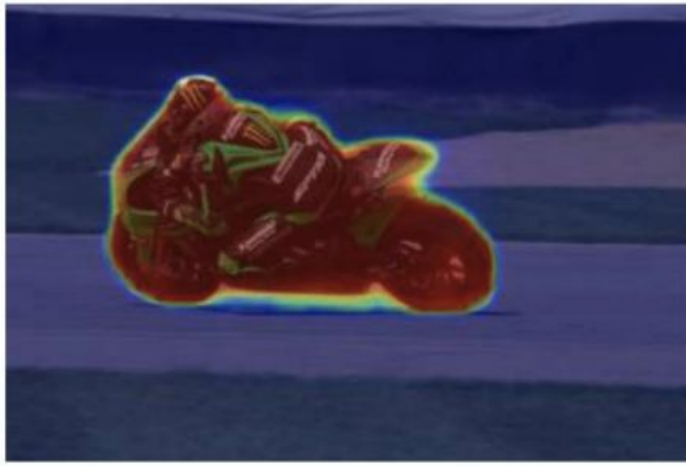


Slide credit: Rodrigo Benenson

Priors

- Boundaries, saliency



Slide credit: Rodrigo Benenson

Priors

- Class Distribution

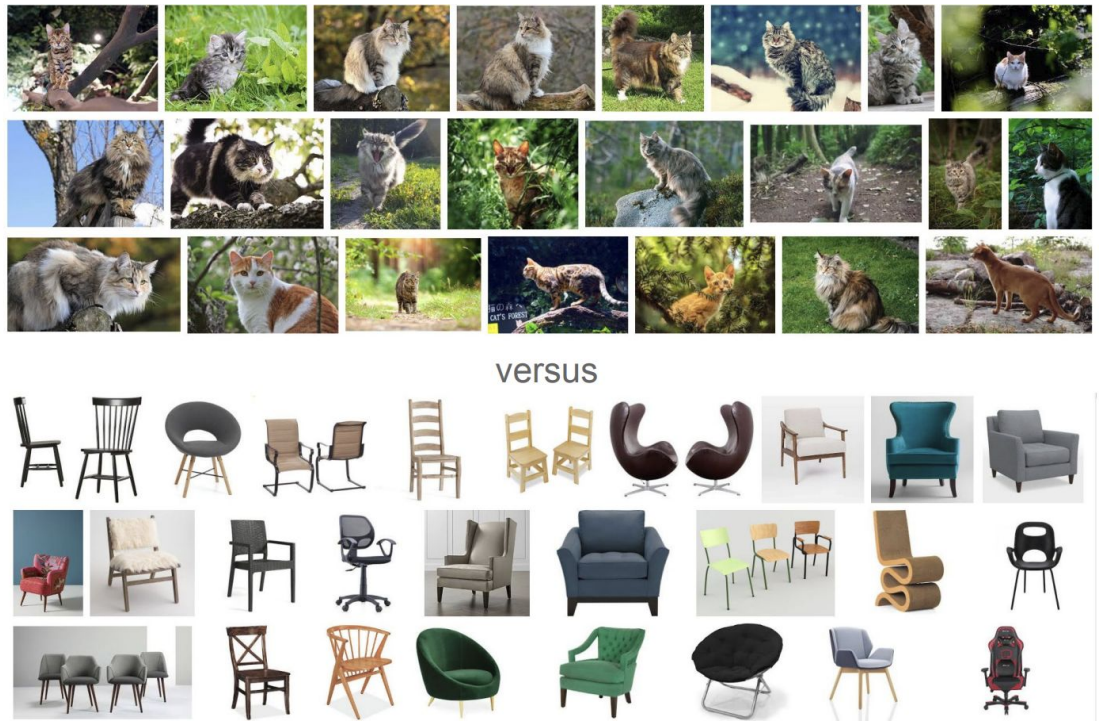


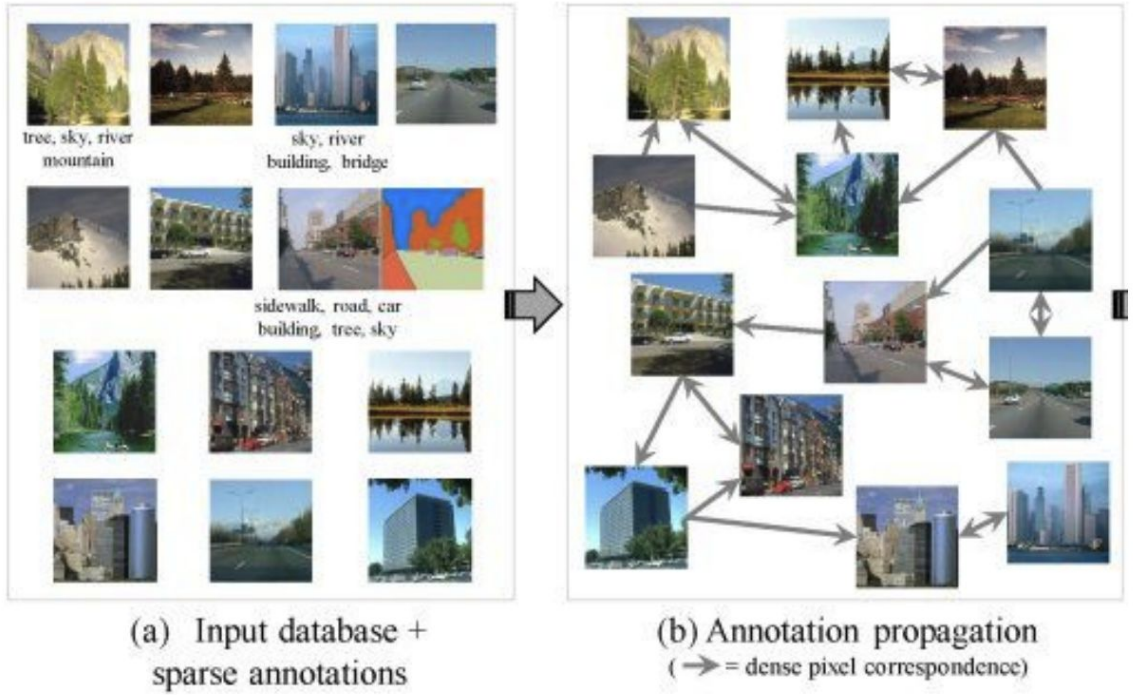
⅔ Cats, ⅓ Cars

Slide credit: Rodrigo Benenson

Priors

- Similarity across images
- Similarity with external images





Slide credit: Rodrigo Benenson

Hints

- Image (video) label
  - Person, cat, door
- Image caption
  - Kattie holds the cat next to the door



[Verbeek CVPR 07, Vezhnevets ICCV 11, Xu CVPR 14, Pinheiro CVPR 15, Pathak ICLR 15, Papandreou ICCV 15, Kolesnikov ECCV 16, Wei CVPR 17, Singh ICCV 17]

Hints

- Image (video) label
  - Person, cat, door
- Image caption
  - Kattie holds the cat next to the door

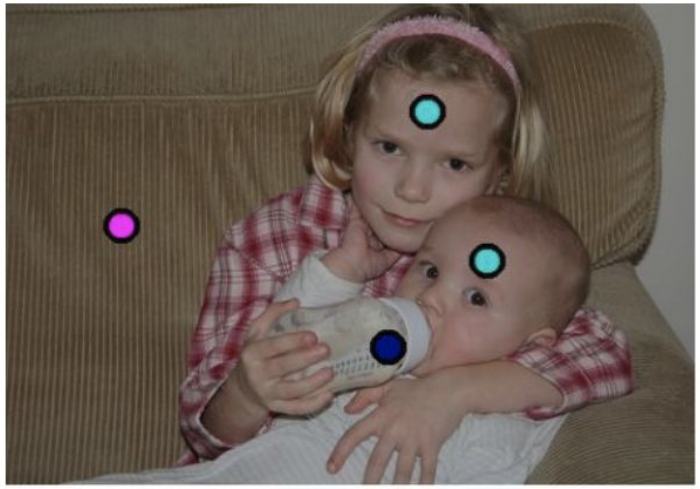


[Verbeek CVPR 07, Vezhnevets ICCV 11, Xu CVPR 14, Pinheiro CVPR 15, Pathak ICLR 15, Papandreou ICCV 15, Kolesnikov ECCV 16, Wei CVPR 17, Singh ICCV 17]

Hints

- Click inside object





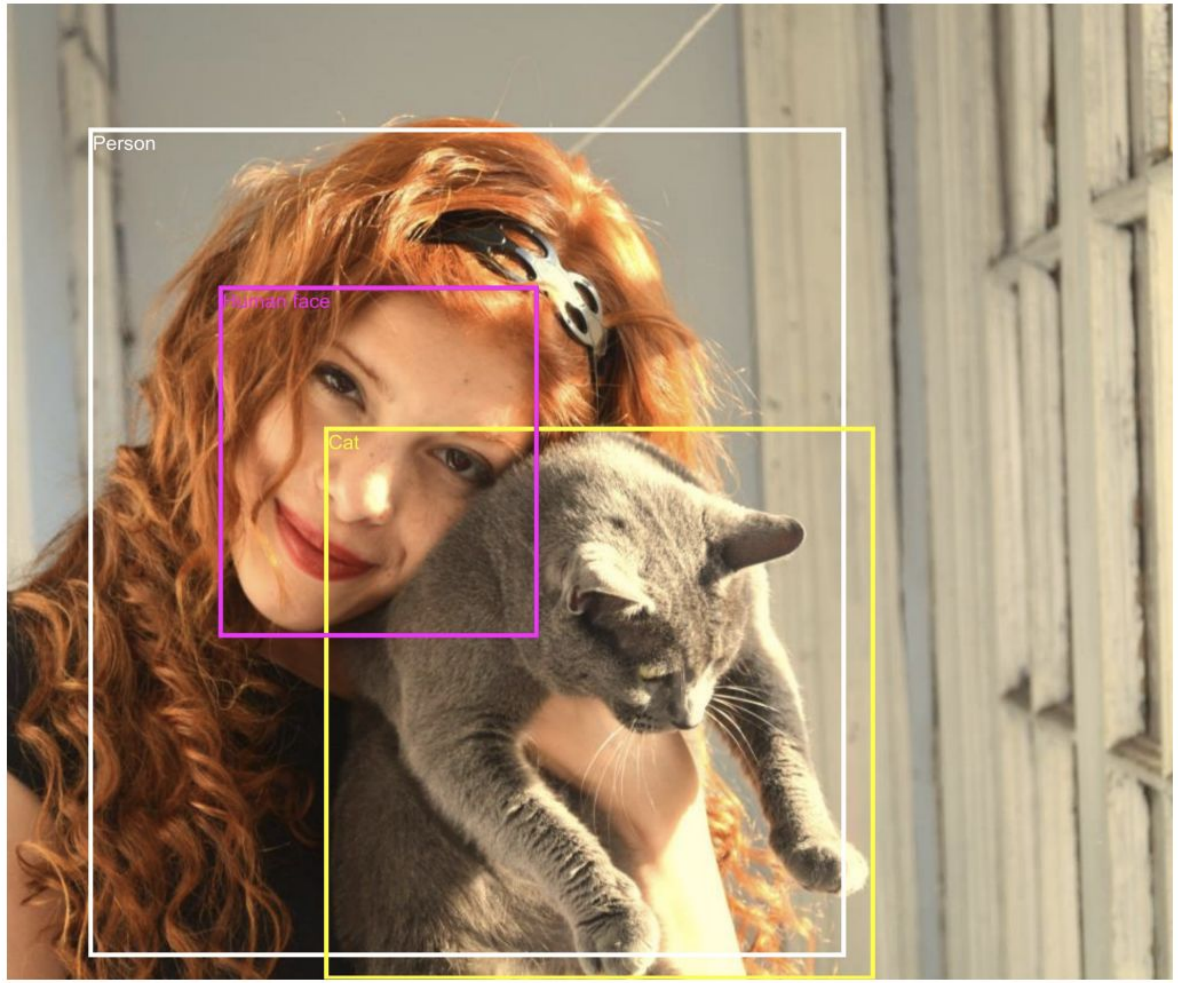




[Verbeek CVPR 07, Vezhnevets ICCV 11, Xu CVPR 14, Pinheiro CVPR 15, Pathak ICLR 15, Papandreou ICCV 15, Kolesnikov ECCV 16, Wei CVPR 17, Singh ICCV 17]

Hints

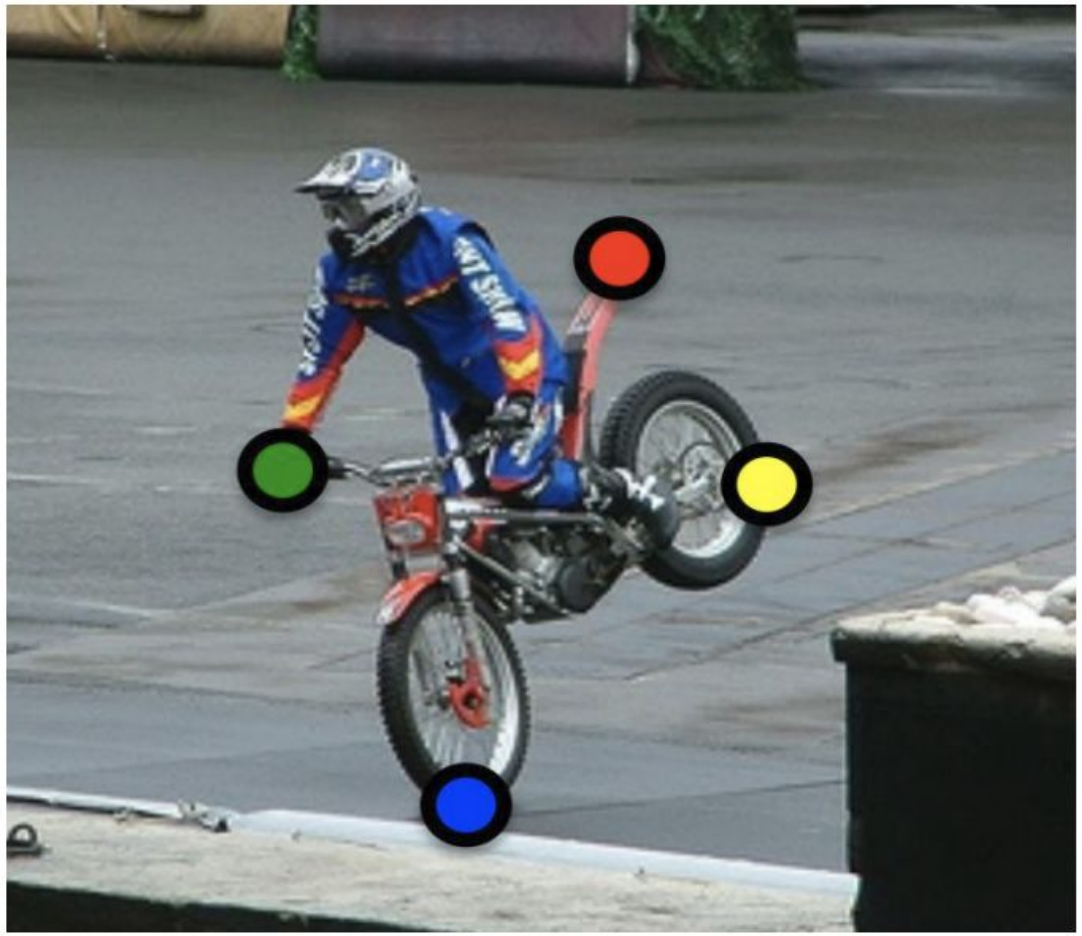
- Object bound boxes

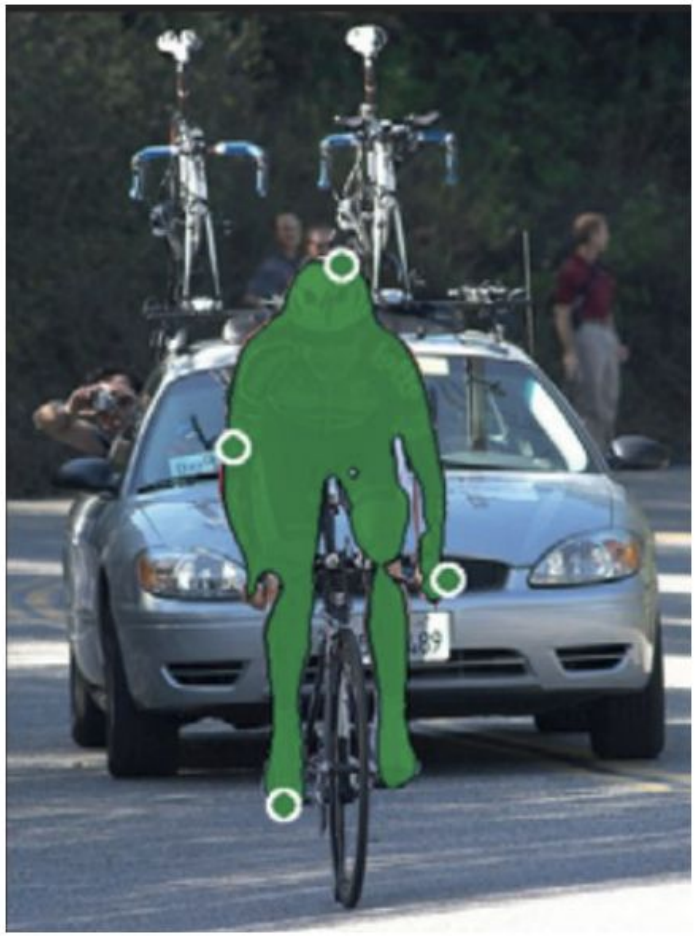


[Verbeek CVPR 07, Vezhnevets ICCV 11, Xu CVPR 14, Pinheiro CVPR 15, Pathak ICLR 15, Papandreou ICCV 15, Kolesnikov ECCV 16, Wei CVPR 17, Singh ICCV 17]

Hints

- Object extreme points



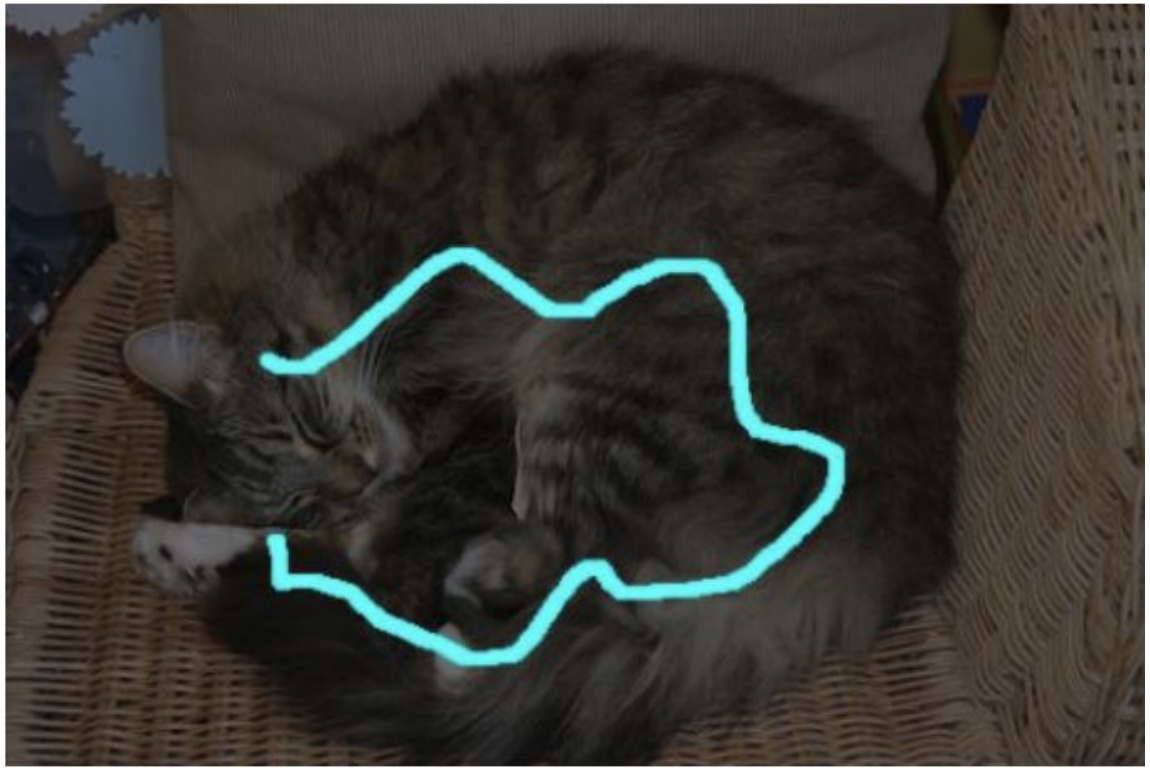


[Verbeek CVPR 07, Vezhnevets ICCV 11, Xu CVPR 14, Pinheiro CVPR 15, Pathak ICLR 15, Papandreou ICCV 15, Kolesnikov ECCV 16, Wei CVPR 17, Singh ICCV 17]

Hints

- Scribbles

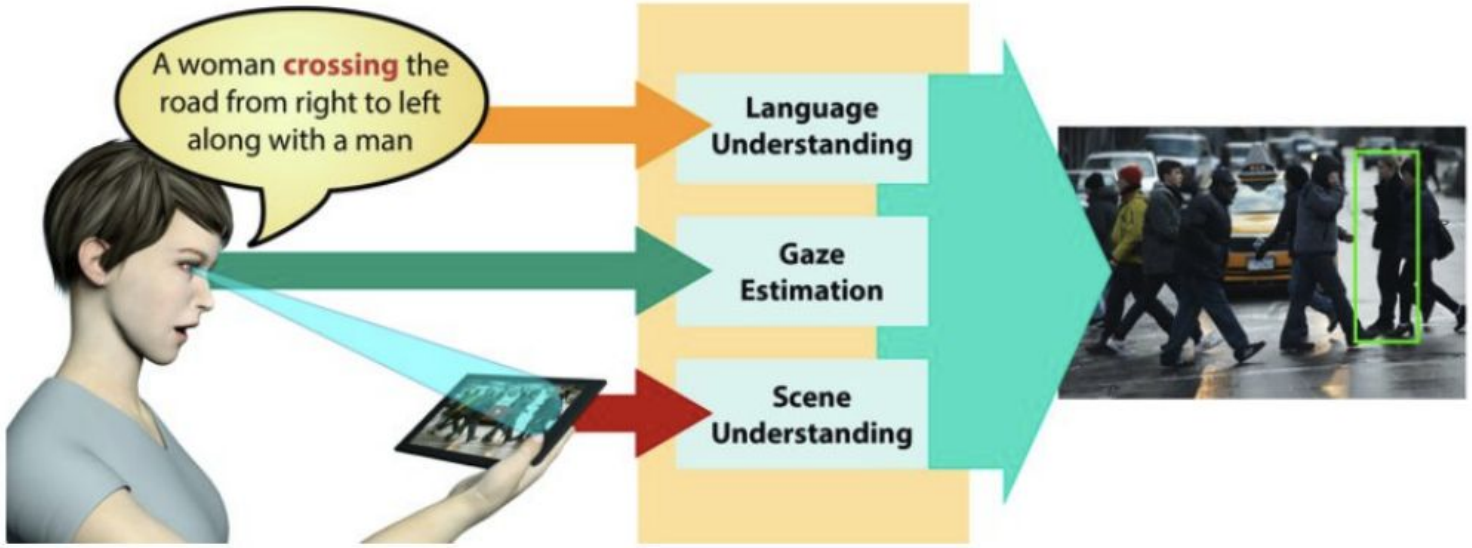


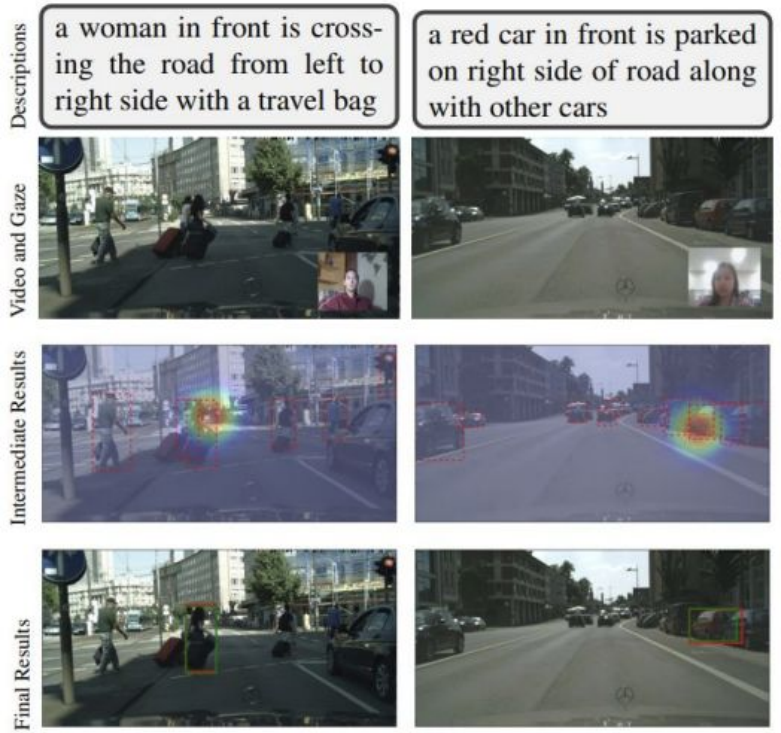


[Verbeek CVPR 07, Vezhnevets ICCV 11, Xu CVPR 14, Pinheiro CVPR 15, Pathak ICLR 15, Papandreou ICCV 15, Kolesnikov ECCV 16, Wei CVPR 17, Singh ICCV 17]

Hints

- Eye gaze





[Verbeek CVPR 07, Vezhnevets ICCV 11, Xu CVPR 14, Pinheiro CVPR 15, Pathak ICLR 15, Papandreou ICCV 15, Kolesnikov ECCV 16, Wei CVPR 17, Singh ICCV 17]

From Image-Level Labels to Pixel-Level Labels

From Image-Level Labels to Pixel-Level Labels



Image-level labels
at training time



Pixel-level labels
at test time

Two Main Approaches

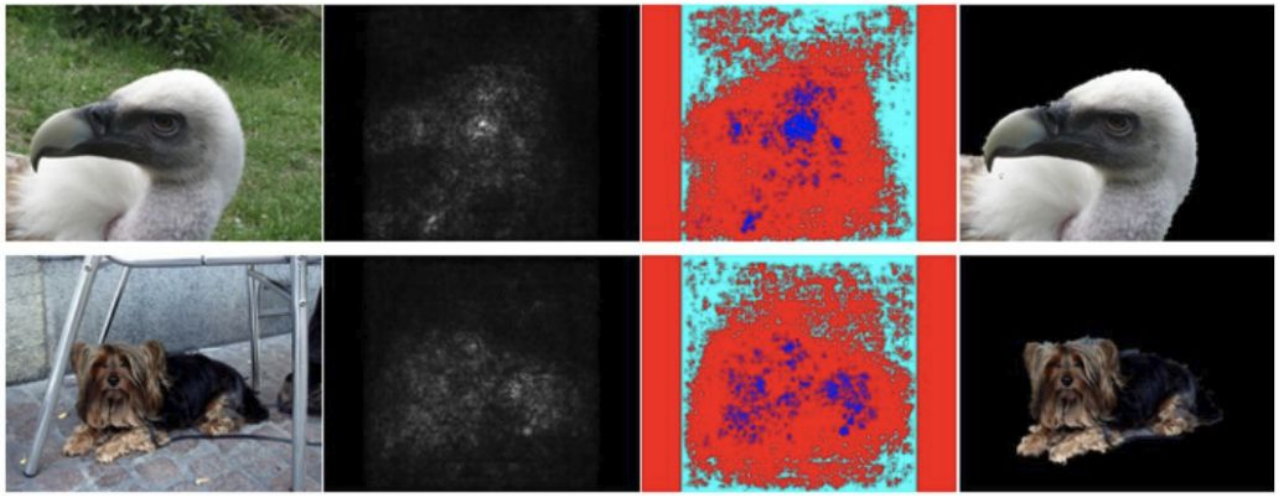
- Gray box
  - Utilize the internal information in the network.
  - In other words, it identifies which areas the model focused on for classification by leveraging the activations or gradients within the CNN.
- White box

Gray Box Approach

- Step 1) Train an ImageNet classifier.
- Step 2) Calculate how much the score changes when the input is slightly modified.

Step 1: Train an image classifier
Step 2: given an image with known label,
compute $\left.\frac{\partial S_c}{\partial I}\right|_{I_0}$

("how much does changes in the input image affect the score?")



- The problem is that it is extremely noisy.
- This is because the gradient measures sensitivity at the level of individual pixels, resulting in a "salt-and-pepper" appearance.

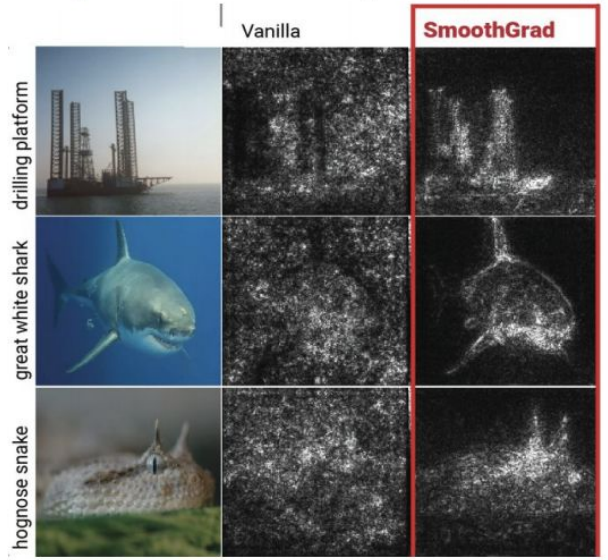
[Simonyan et. al 2014]

Gray Box Approach

- Step 1) Training an Image Classifier
  - Train a standard CNN classifier.
- Step 2) Add small random noise to the images
- Step 3) Calculate the gradient for each
- Step 4) Average

This slide explains SmoothGrad.
The key point is that calculating the gradient multiple times and averaging it removes noise and yields a more meaningful saliency map.

Step 1: Train an image classifier
Step 2: **Add small random noise** to image
Step 3: given known label,
compute $\left.\frac{\partial S_c}{\partial I}\right|_{I_0}$
("how much does changes in the input image affect the score?")
Step 4: **Repeat** 50 times step 2&3, **average result**



(no quantitative experiments either)

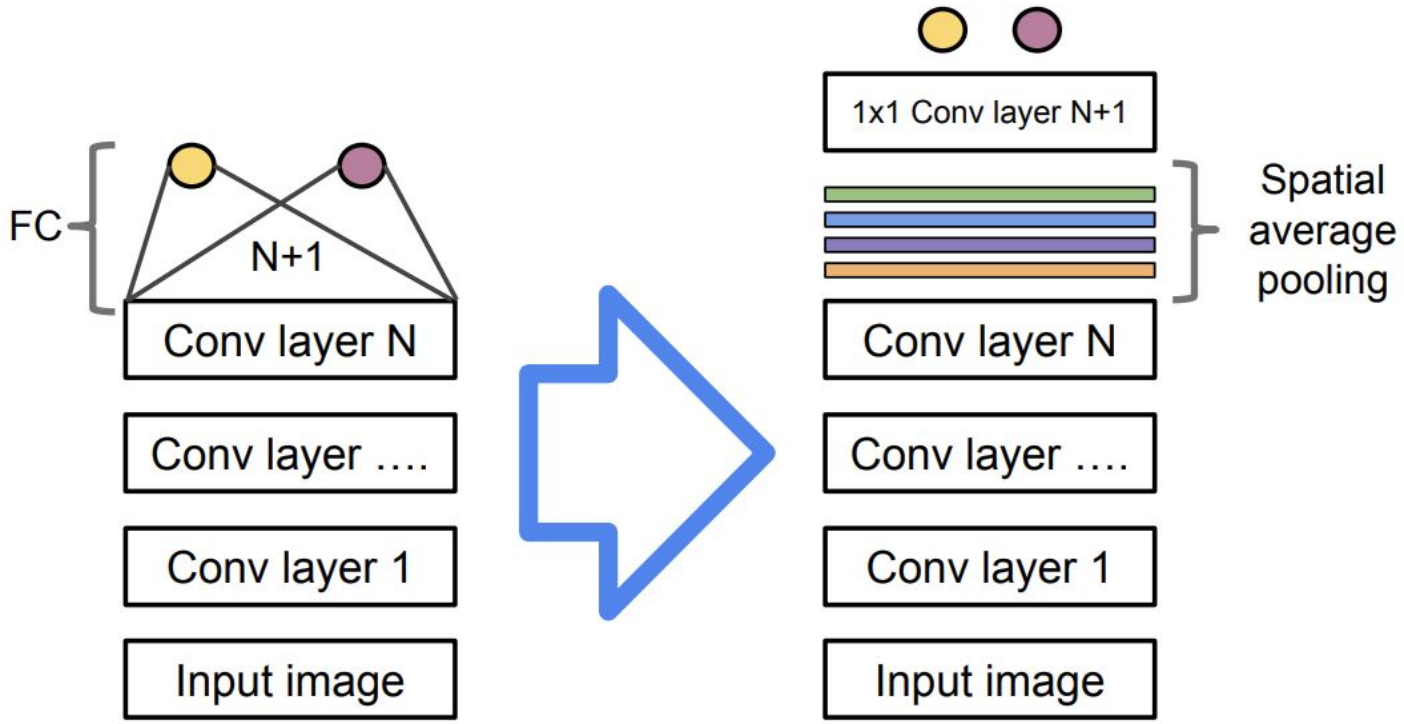
[Smilkov et al. 2017]

Gray Box Approach

- Key issue: Asking apples from the oranges tree **Classifier was not built nor trained to segment objects.**
- Gradient-based saliency involves forcing localization using a classifier. Since the classifier was not originally trained for localization, it suffers from a **fundamental limitation**: it highlights only the specific regions necessary for classification rather than the entire object.

White Box Approach

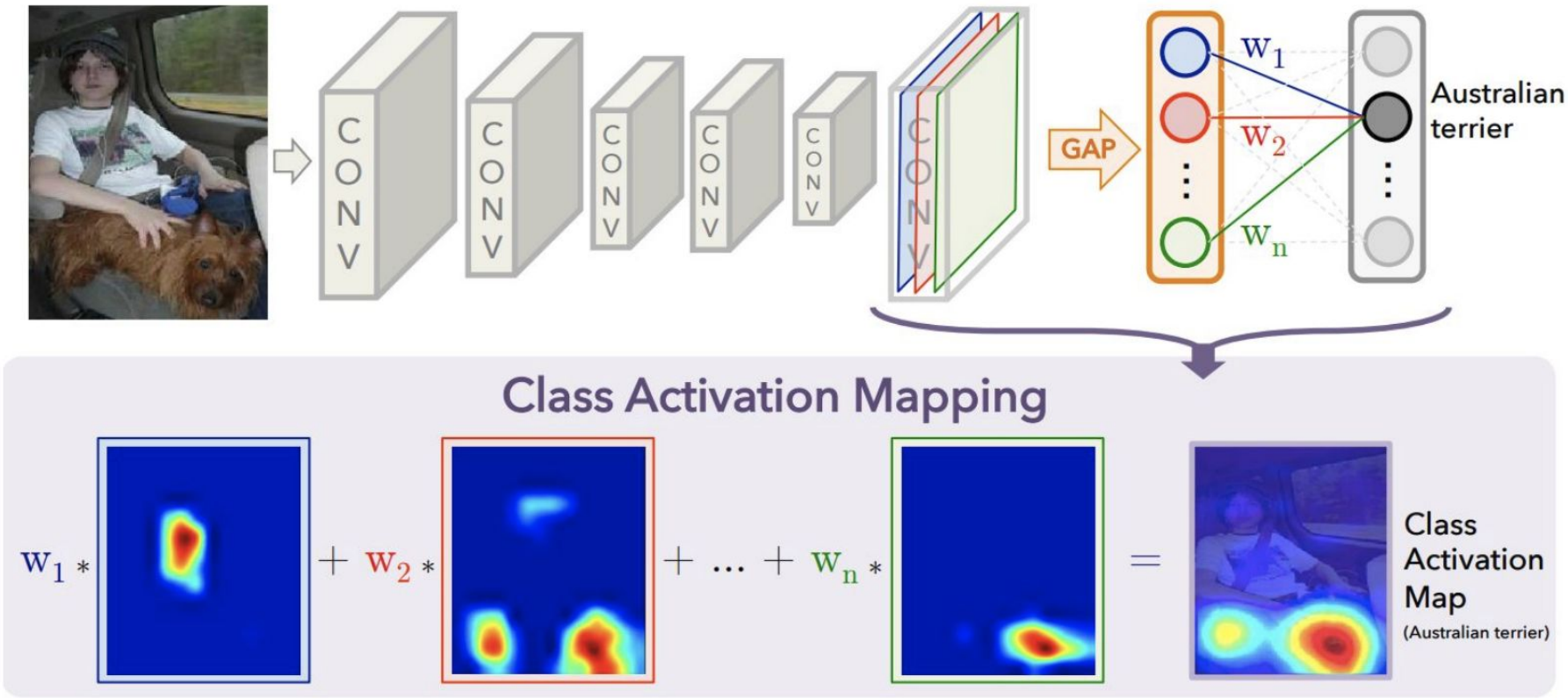
- Step 1) eliminate the FC layer
- Step 2) add GAP (Global Average Pooling)
- Step 3) add a 1×1 convolution that performs a linear combination of the features



[Zhou et al. 2016]

White Box Approach

- CAM (Class Activation Mapping)



since the network itself is designed to preserve spatial information, activation maps can be obtained directly without the need for separate gradient calculations.

[Zhou et al. 2016]

Designed to Generate Per-Class Maps

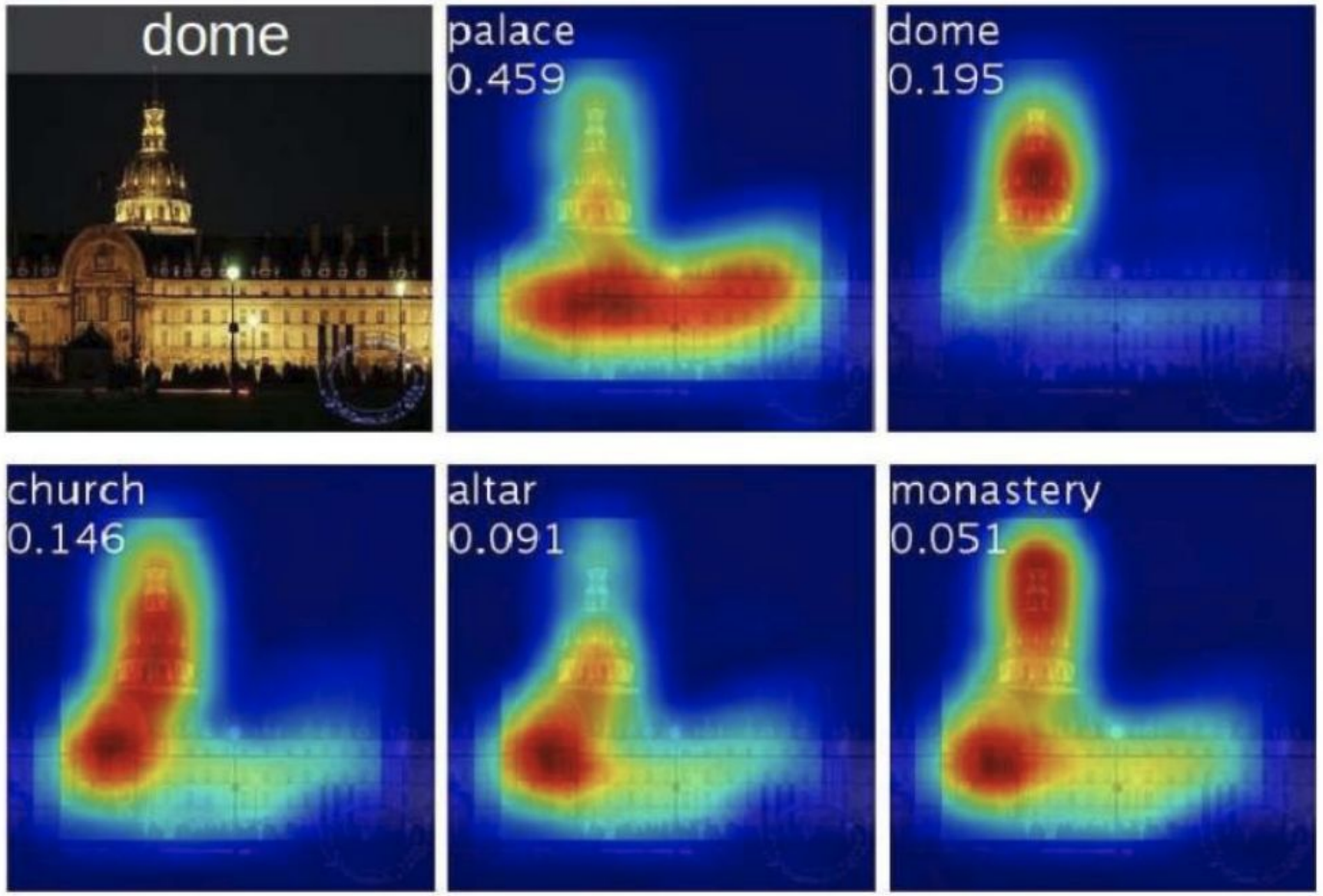
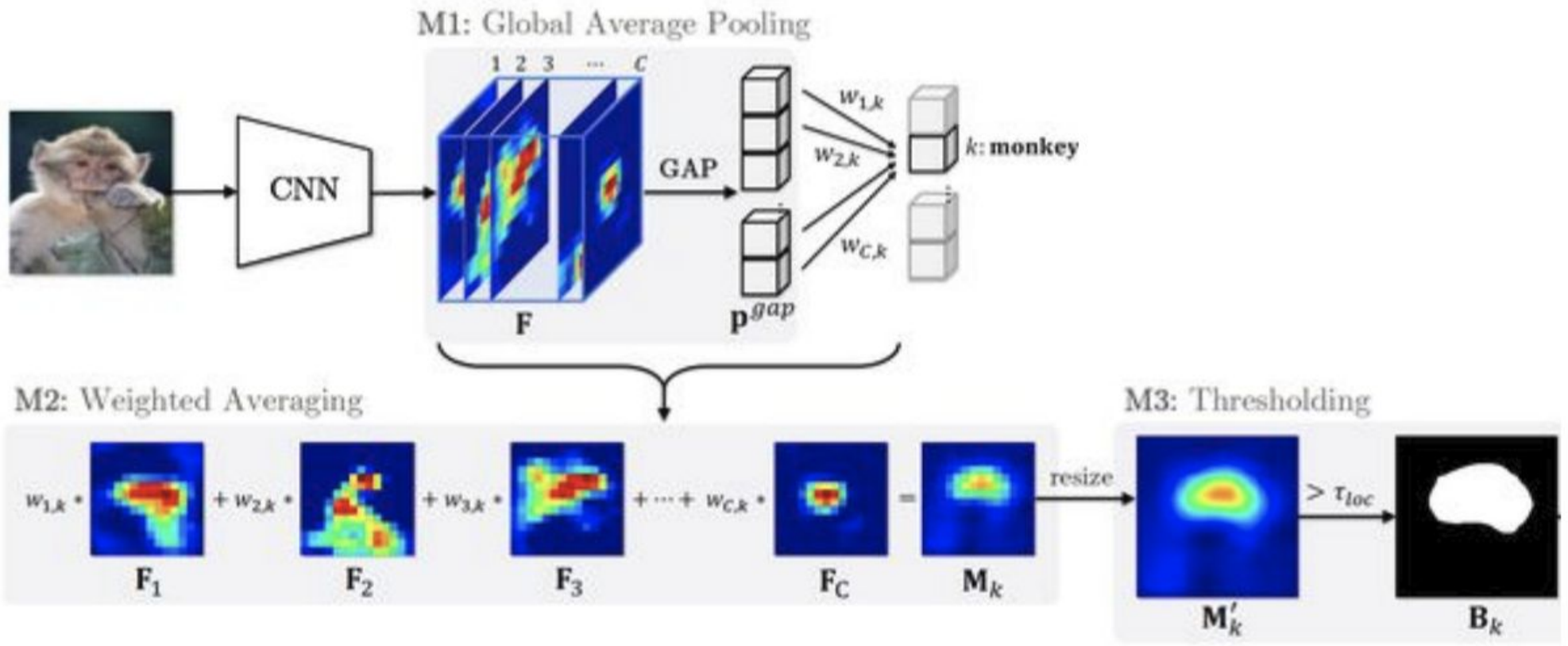


Figure 4. Examples of the CAMs generated from the top 5 predicted categories for the given image with ground-truth as dome.

[Zhou et al. 2016]

The Class Maps are Thresholded, to Generate Training Data

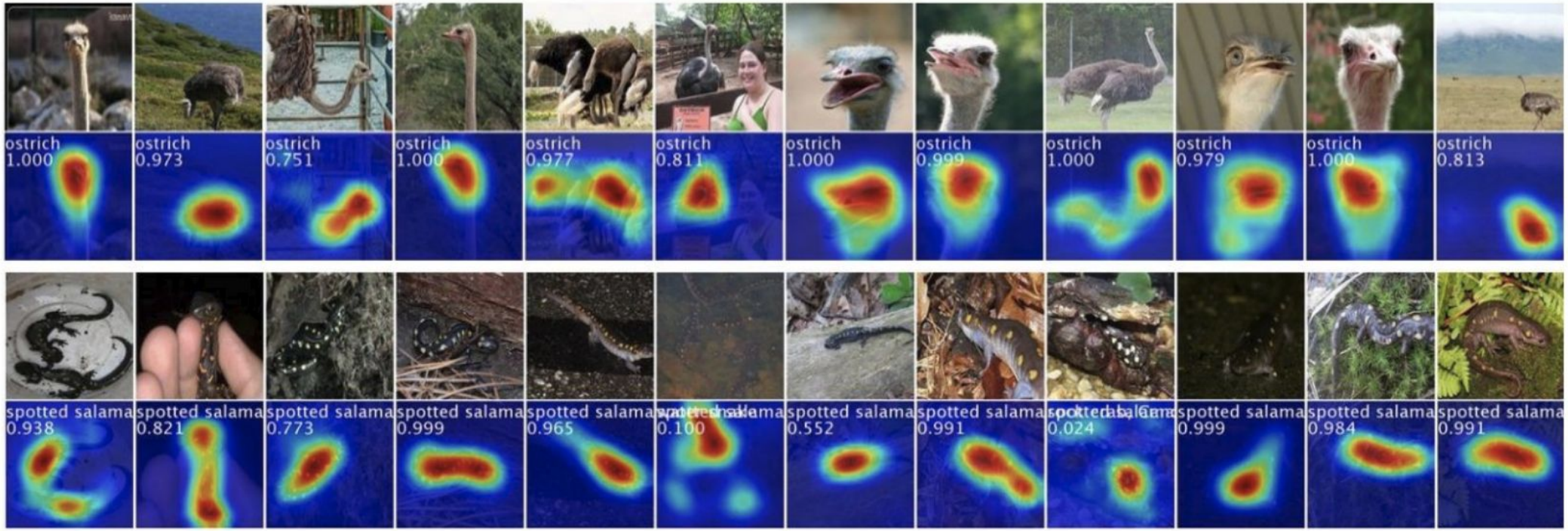


High value scores → Foreground areas
Low value scores → Background areas
(This confident regions are commonly named "seeds")

[Zhou et al. 2016]

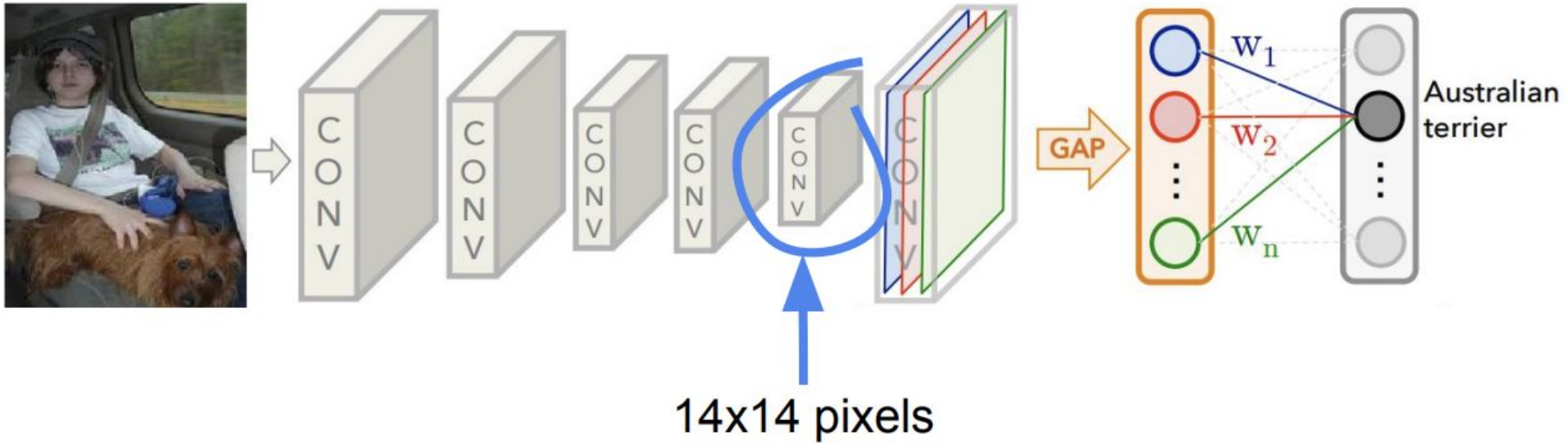
Good, but not good enough

- 1. Resulting masks are not sharp
- 2. Focused on discriminative area only



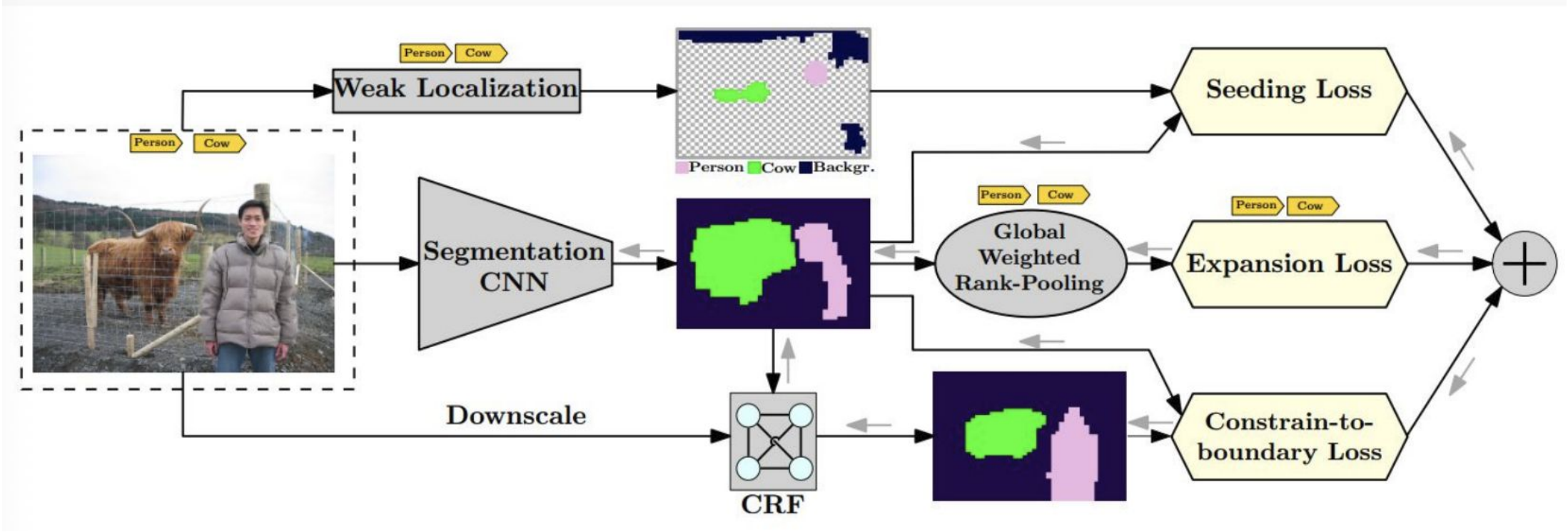
Good, but not good enough

- 1. Resulting masks are not sharp
  - ⇢ Increase top layers resolution
- 2. Focused on discriminative area only



Good, but not good enough

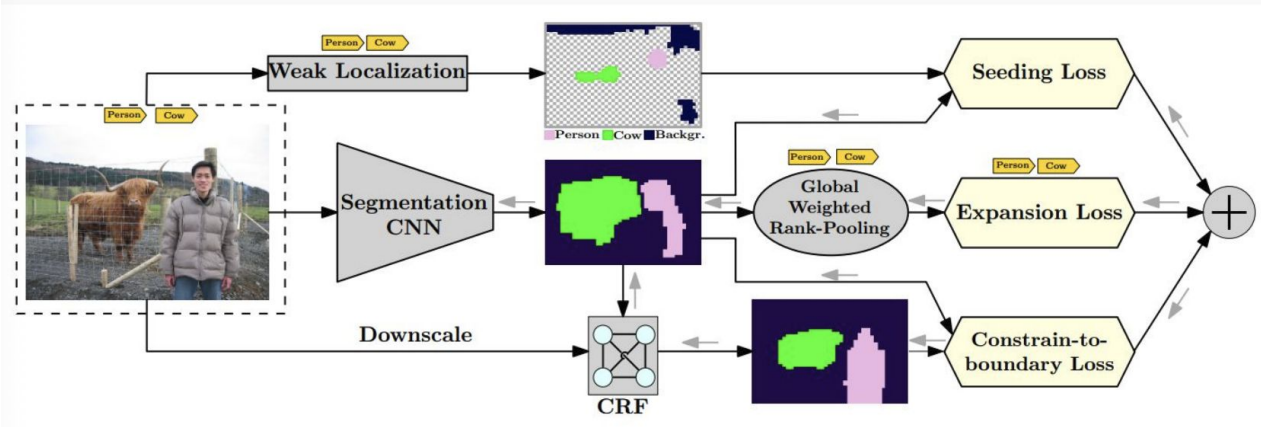
- 1. Resulting masks are not sharp
  - ⇢ Use boundary-aware propagation
- 2. Focused on discriminative area only



[Kolesnikov & Lampert ECCV 2016]

Good, but not good enough

- CAM limitations
  - Activation maps are small and coarse
  - Focus only on discriminative regions
  - Poor object boundaries
- Three complementary losses
  - 1. Seeding Loss
    - Use CAM as reliable foreground seeds
    - Preserve highly confident object regions
  - 2. Expansion Loss
    - Encourage predictions to expand beyond the seeds
    - Use image-level labels via Global Weighted Rank Pooling (GWRP)
  - 3. Constrain-to-boundary Loss
    - Refine predictions using CRF
    - Encourage masks to align with image boundaries

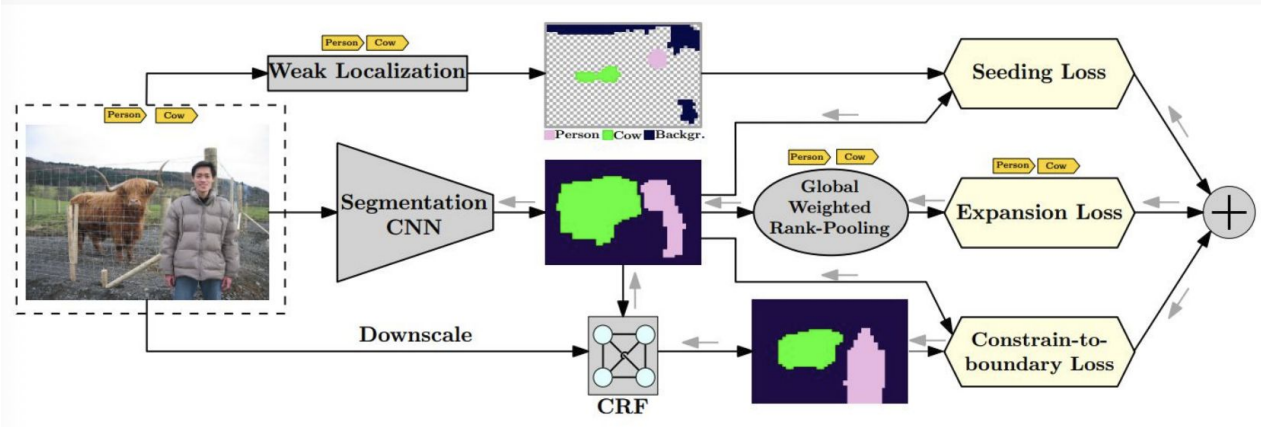


SEC (Seed, Expand, Constrain)
Seed: Preserve reliable CAM activations.
Expand: Grow seeds to cover the entire object using image-level supervision.
Constrain: Refine object boundaries with CRF-based regularization.

[Kolesnikov & Lampert ECCV 2016]

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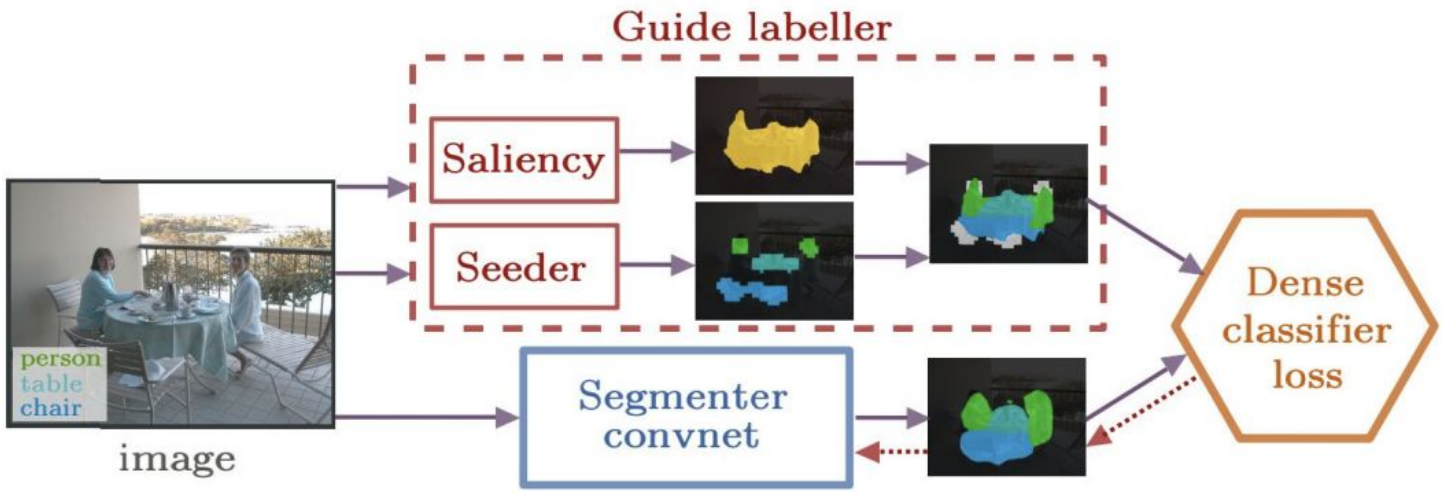
- 1. Resulting masks are not sharp
  - ⇢ Use transfer learning
- 2. Focused on discriminative area only



image labels



saliency



"We could also use saliency cues"

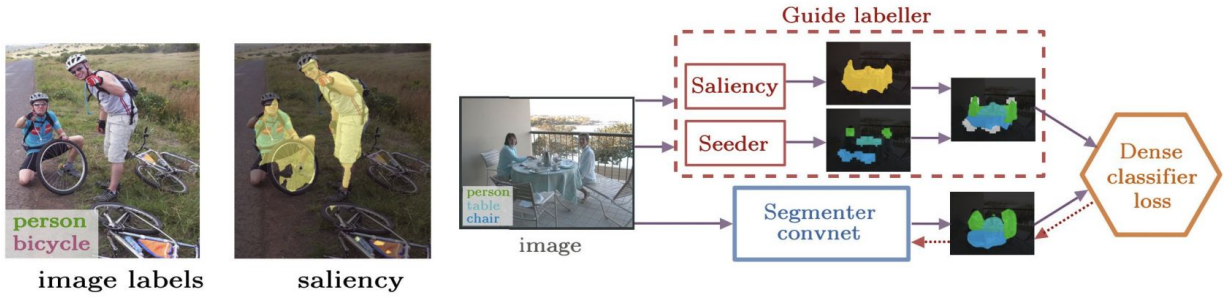
[Oh et al. CVPR 2017]

Good, but not good enough

- Observation
  - CAM
    - Provides class-specific localization
    - Covers only discriminative regions
  - Saliency
    - Provides complete object extent
    - Class-agnostic foreground estimation
  - Combine CAM and saliency to generate class-specific pseudo masks

Saliency-guided Weak Supervision

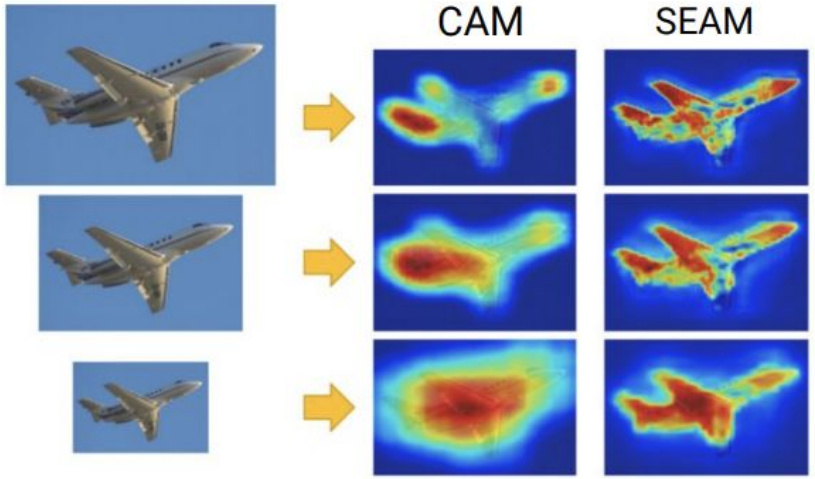
- **Seeder: Generate class-specific seeds from CAM.**
- **Saliency: Estimate foreground regions.**
- **Guide Labeller: Fuse both cues into pseudo masks.**
- **Segmentation CNN: Learn dense pixel-wise predictions.**

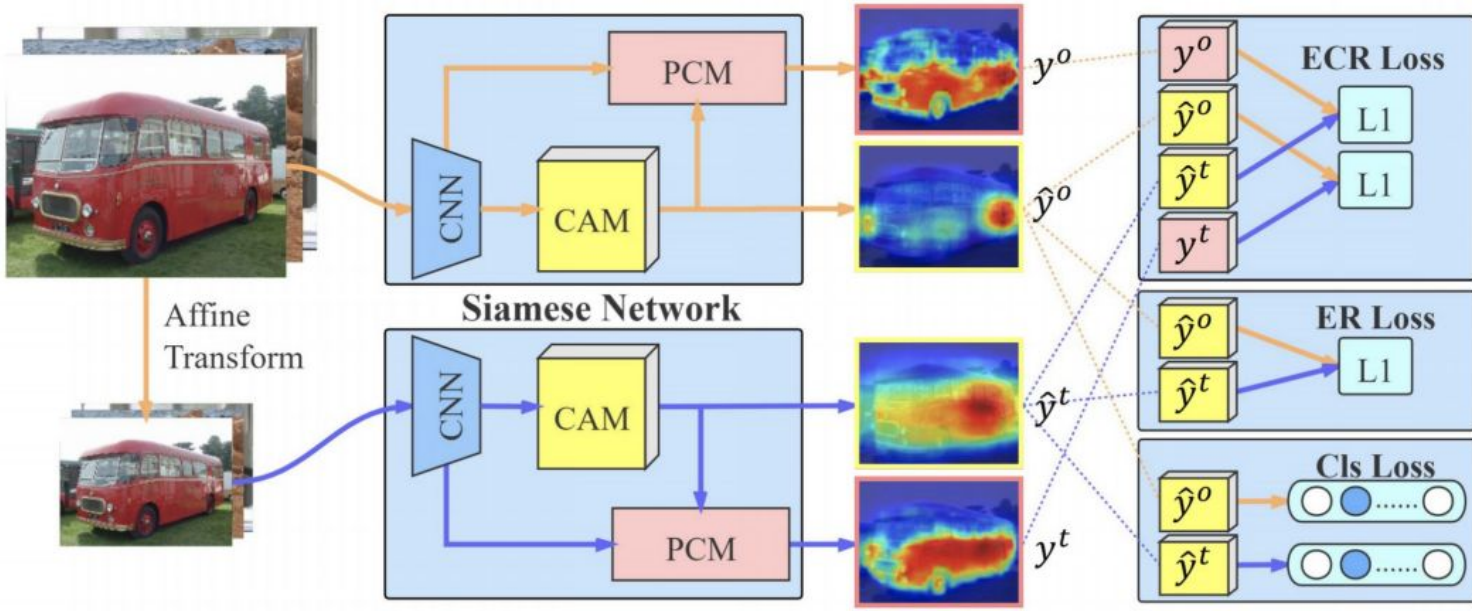


[Oh et al. CVPR 2017]

Good, but not good enough

- 1. Resulting masks are not sharp
  - ⇢ Enforce equivariance
- 2. Focused on discriminative area only



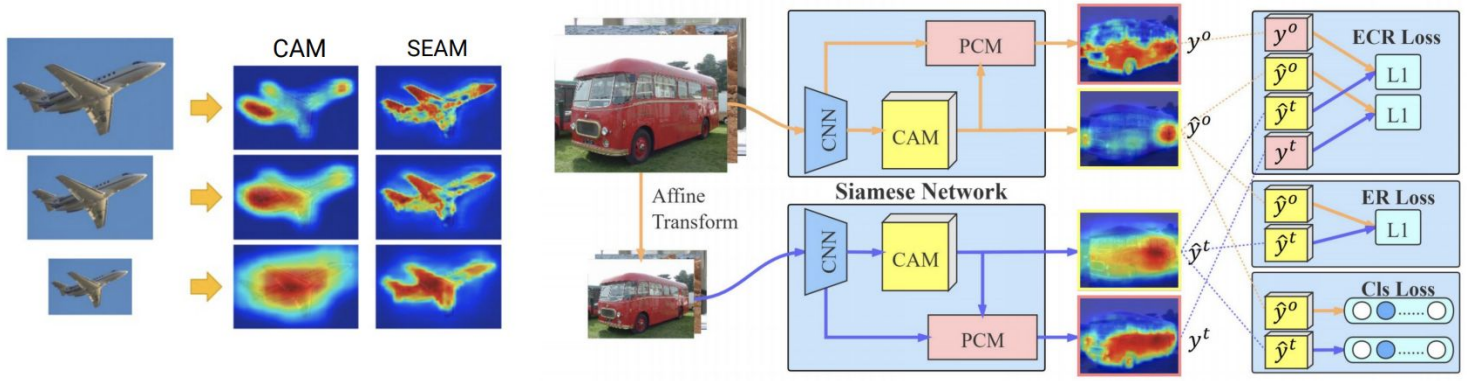


"Activation maps should be equivariant to the input"

[Wang et al. CVPR 2020]

Good, but not good enough

- CAM Limitation
  - CAMs are often coarse and inconsistent
  - Small input transformations can produce different activation maps
- Key idea: Enforce Equivariance
  - Apply geometric transformations (e.g., affine transform)
  - Predict CAMs for both original and transformed images
  - Enforce consistency after compensating for the transformation



PCM (Pixel Correlation Module) is a module that propagates CAM activations to other pixels within the same object by utilizing pixel-wise affinity among CNN features.

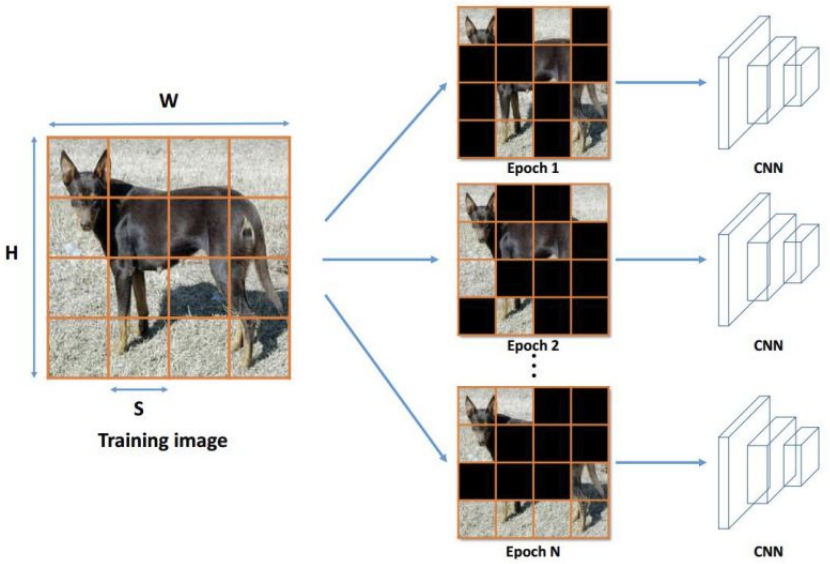
Training objective

- **Classification Loss: Preserve image-level classification performance**
- **Equivariant Regularization (ER) Loss: Encourage transformation-consistent CAMs**
- **Cross-view Consistency (ECR) Loss: Align CAMs across different views**

[Wang et al. CVPR 2020]

Good, but not good enough

- 1. Resulting masks are not sharp
- 2. Focused on discriminative area only
  - Problem
    - CAM activates only the most discriminative regions.
  - Solution (Hide-and-Seek)
    - Randomly hide image patches during training.
    - Encourage learning from the remaining visible parts.
    - Generate more complete localization maps.
  - Effect
    - The classifier is forced to discover other visible object parts.
    - CAM gradually covers a larger portion of the object.

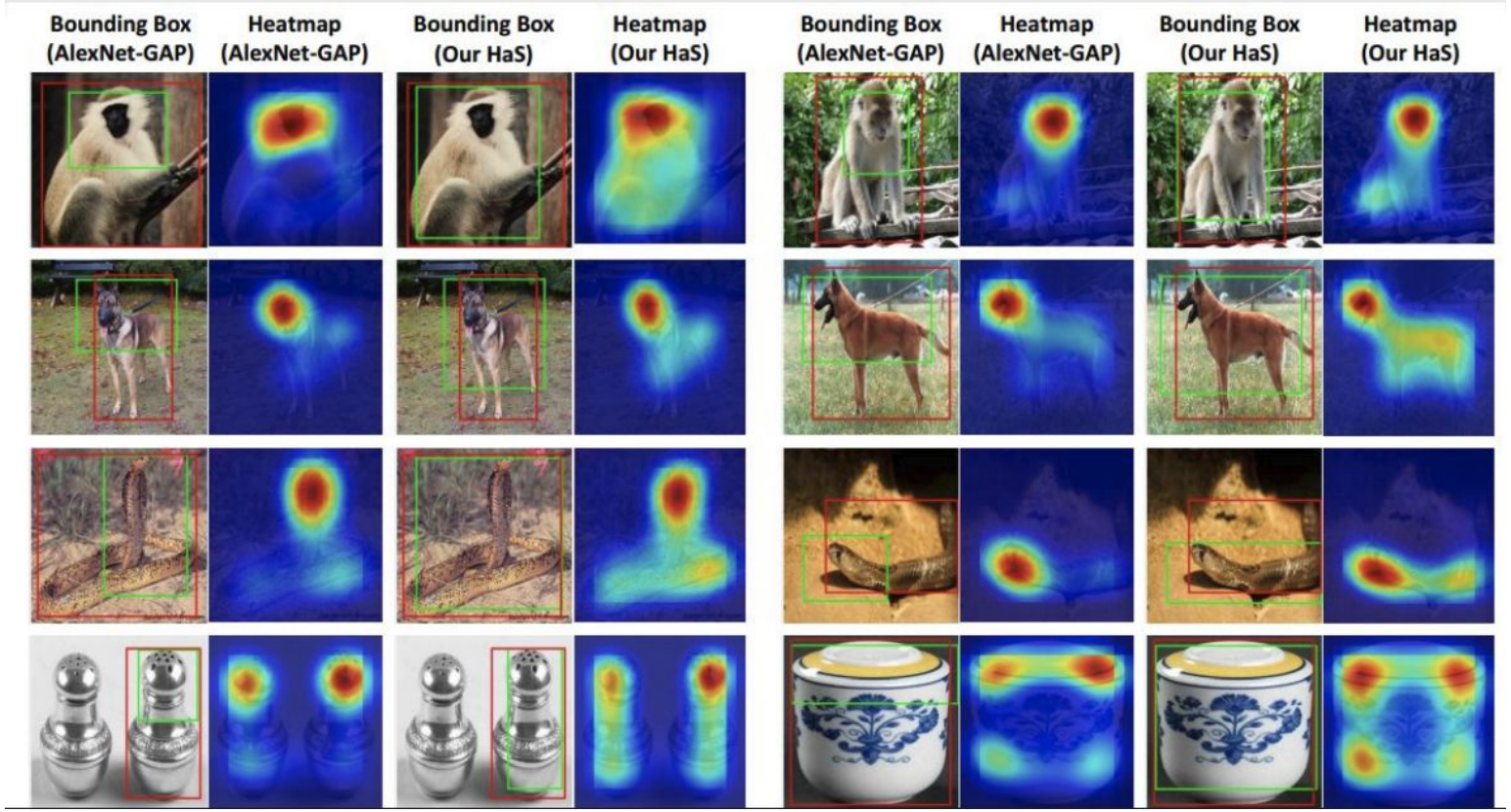


Random hiding encourages the network to discover complementary object parts instead of relying on a single discriminative region.

[Wang et al. CVPR 2020]

Good, but not good enough

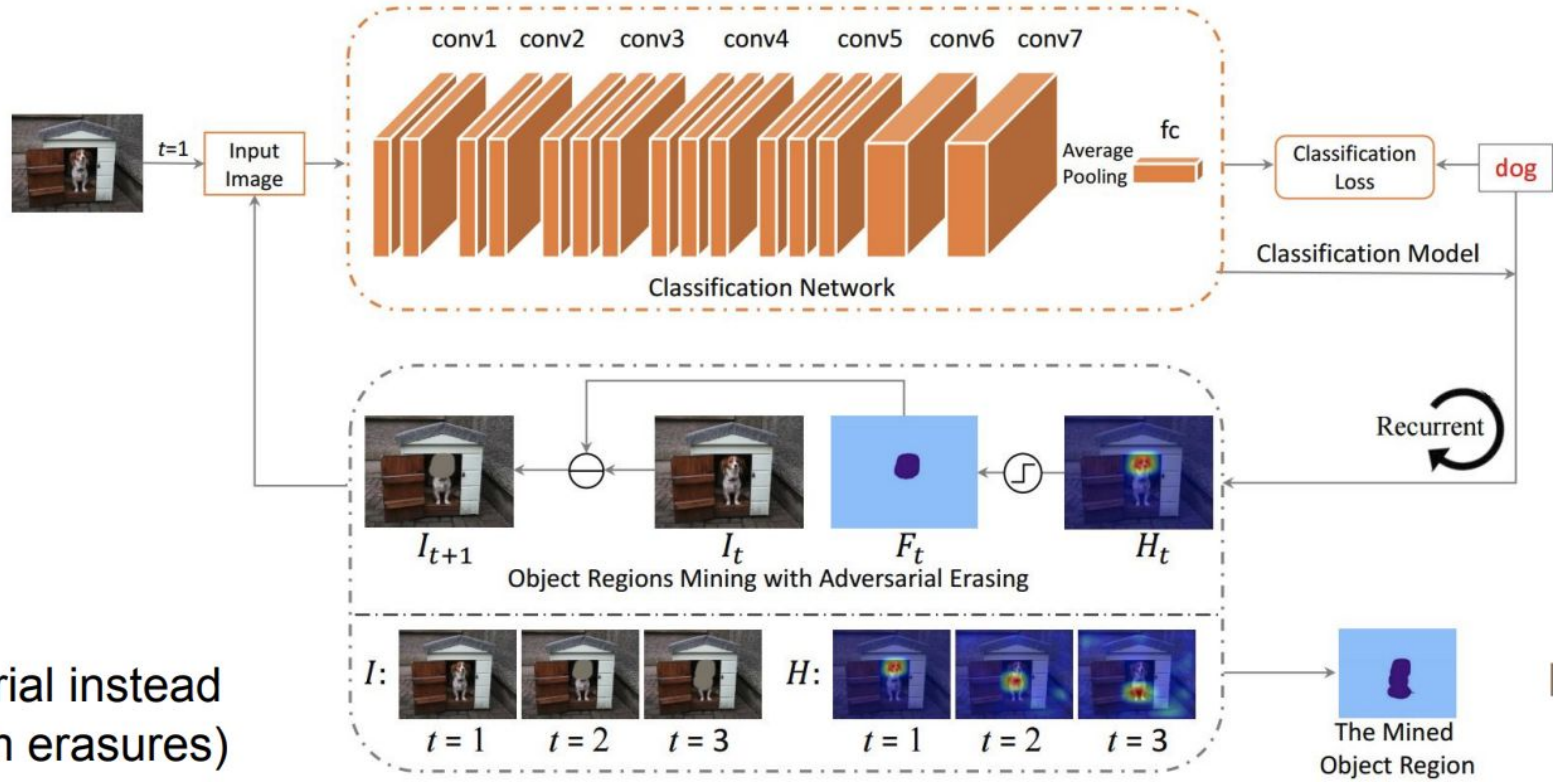
- 1. Resulting masks are not sharp
- 2. Focused on discriminative area only
  - ⇢ Let us play peek-a-boo!



[Wang et al. CVPR 2020]

Good, but not good enough

- 1. Resulting masks are not sharp
- 2. Focused on discriminative area only
  - ⇢ Let us play adversarial peek-a-boo!



(adversarial instead of random erasures)

[Wei et al. 2017]

Good, but not good enough

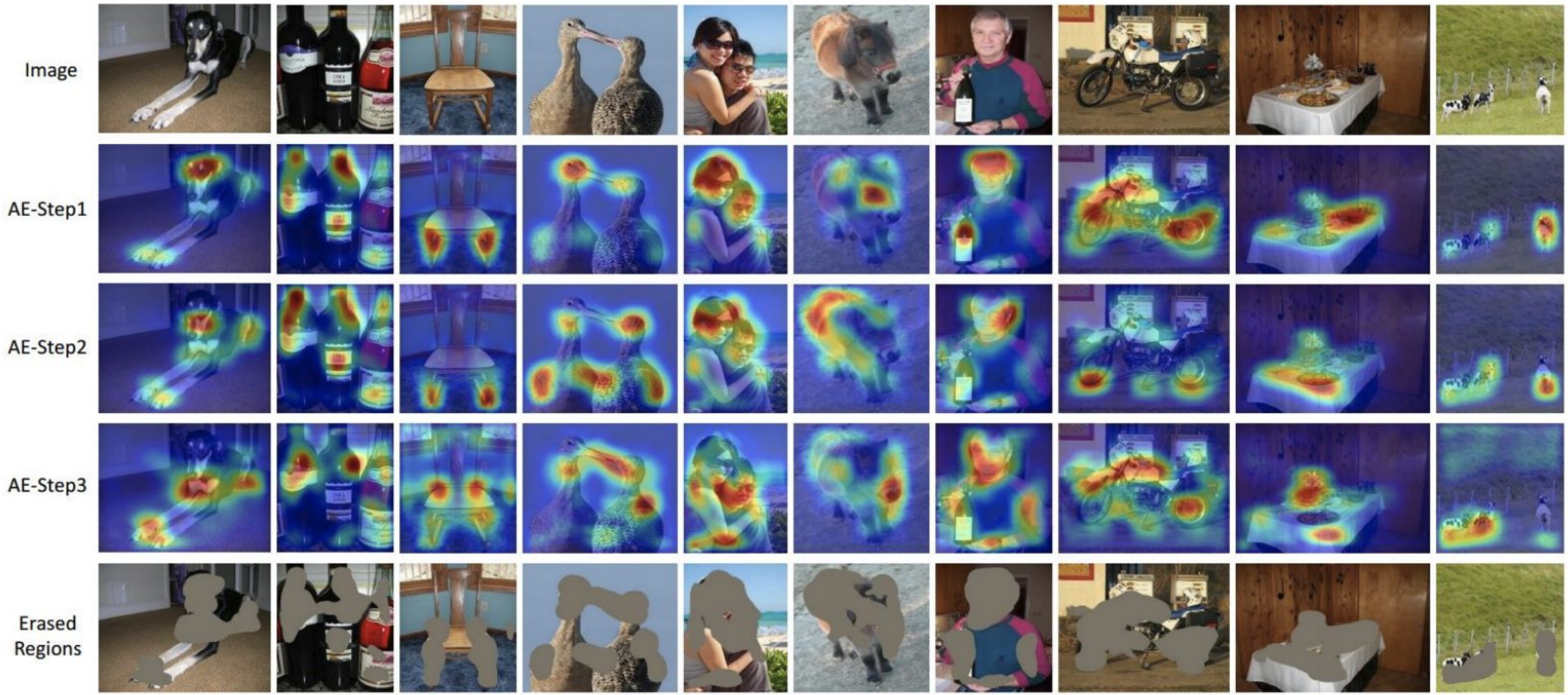


Figure 6. Examples of mined object regions produced by the proposed adversarial erasing approach. The second to fourth rows show the produced heatmaps, where the discriminative regions are highlighted. The images with erased regions are shown in the last row in gray.

[Wei et al. 2017]

Good, but not good enough

- 1. Resulting masks are not sharp
- 2. Focused on discriminative area only
  - Let us play adversarial peek-a-boo!

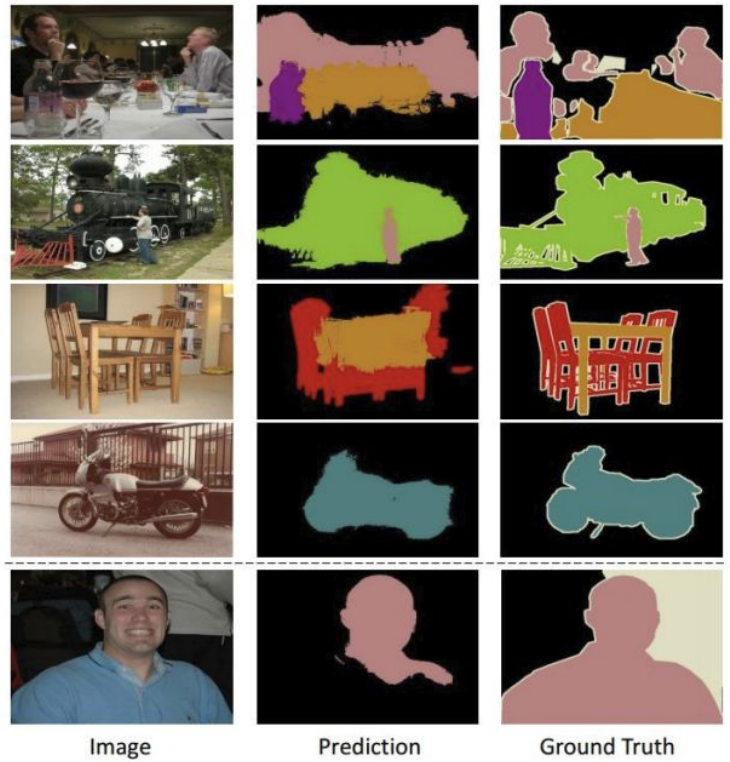


Figure 4. Qualitative segmentation results on the VOC 2012 *val* set. One failure case is shown in the last row.

Table 2. Comparison of weakly-supervised semantic segmentation methods on VOC 2012 *test* set.

Methods	Training Set	mIoU
Supervision: Box		
WSSL (ICCV 2015) [16]	10K	62.2
BoxSup (ICCV 2015) [3]	10K	64.2
Supervision: Image-level Labels (* indicates methods implicitly use pixel-level supervision)		
MIL-seg* (CVPR 2015) [19]	700K	40.6
SN_B* (PR 2016) [28]	10K	43.2
TransferNet* (CVPR 2016) [7]	70K	51.2
AF-MCG* (ECCV 2016) [20]	10K	55.5
Supervision: Image-level Labels		
MIL-FCN (ICLR 2015) [18]	10K	24.9
CCNN (ICCV 2015) [17]	10K	35.6
MIL-sppxl (CVPR 2015) [19]	700K	35.8
MIL-bb (CVPR 2015) [19]	700K	37.0
EM-Adapt (ICCV 2015) [16]	10K	39.6
DCSM (ECCV 2016) [24]	10K	45.1
BFBP (ECCV 2016) [23]	10K	48.0
STC (PAMI 2016) [29]	50K	51.2
SEC (ECCV 2016) [10]	10K	51.7
AF-SS (ECCV 2016) [20]	10K	52.7
Supervision: Image-level Labels		
AE-PSL (ours)	10K	**55.7**

Takeaways

- Training a classifier, and hoping to get segmentations out of it ✗
- Using spatial continuity priors will help, but to a limit ~
- The training can be modified to promote covering the object extent ✔
- With careful changes it is possible to obtain significant improvements ~
- ⇒ How far can we go from image-level to pixel-level labels ?

Webly Supervised Learning

Webly Supervised Learning

- A learning method that uses data collected from the web as supervision.
  - Collect training images directly from web search engines.
  - Enables large-scale learning without manual annotation.
  - Labels are noisy, ambiguous, and incomplete.
  - Learning must be robust to noisy supervision.

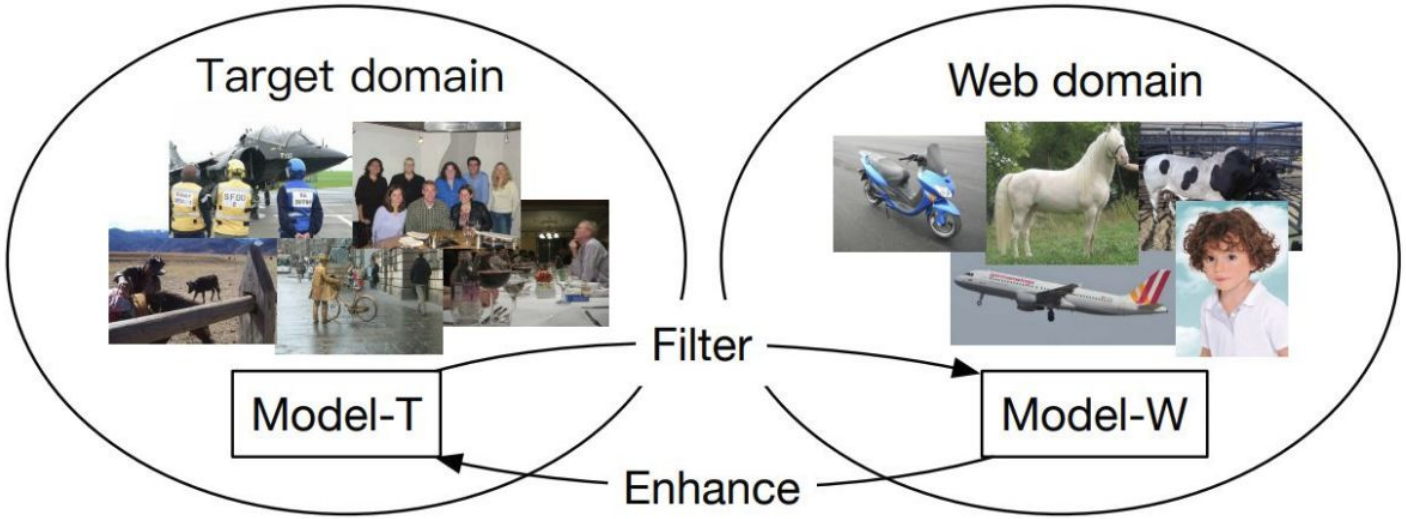
Webly supervised learning leverages freely available web images as supervision, trading annotation cost for label noise.



[Shen CVPR 18, Jin CVPR 17, Chen ICCV 15]

Webly Supervised Learning

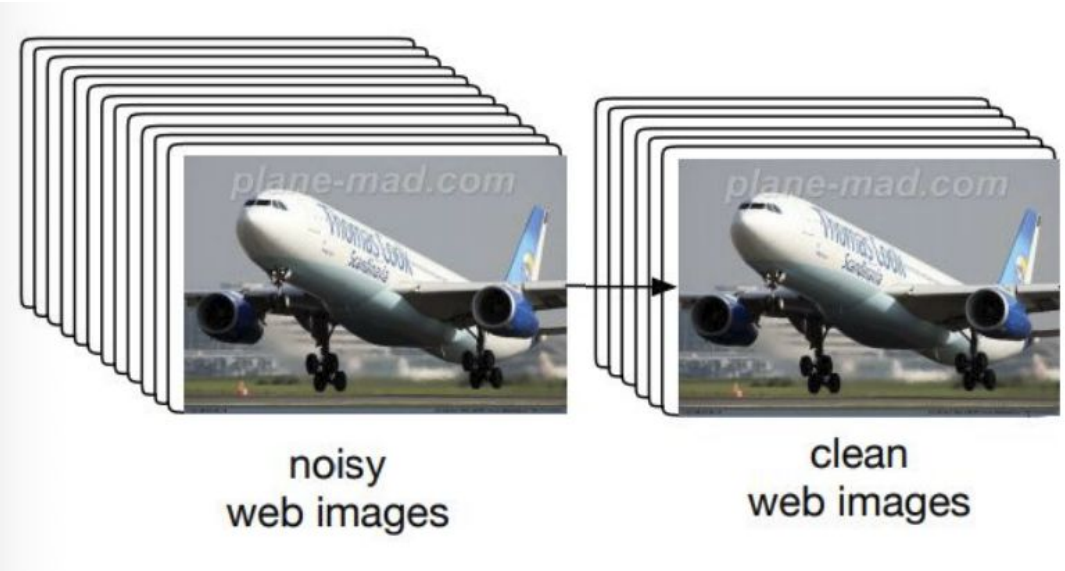
- Pipeline
  - **1. Train an image classifier (Model-T) on clean image-level labels.**
  - 2. Use Model-T to filter noisy web images.
  - 3. Train Web-SEC (pixel labeler) using the filtered web images.
  - 4. Apply Web-SEC to generate pixel labels for the target images.
  - 5. Use Model-T to produce class activation maps (CAMs) on the target images.
  - 6. Fuse the two label sources and train the final segmentation model.



[Shen CVPR 18]

Webly Supervised Learning

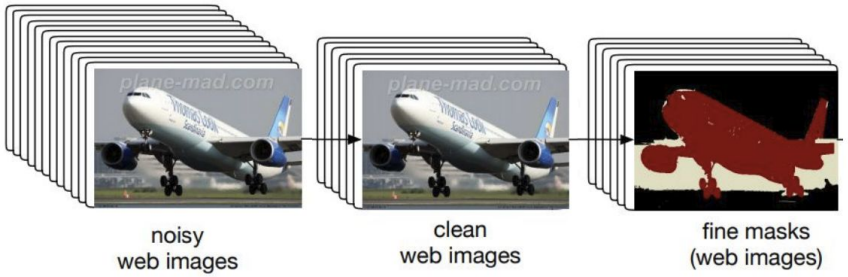
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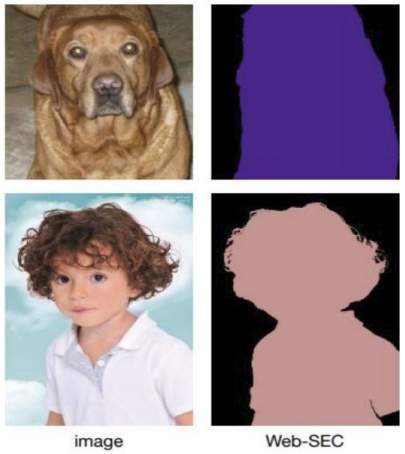


[Shen CVPR 18]

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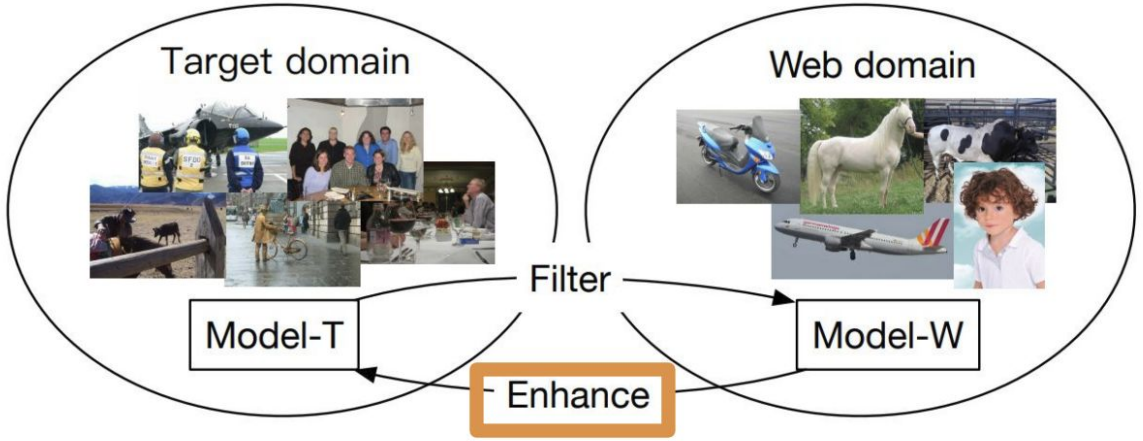


- Unlike CAM-based methods, Web-SEC actually learns a pixel classifier.
- The key observation is that web images are much cleaner—typically containing a single centered object—so approximate segmentation masks can be generated automatically.
- These masks are then used to train a pixel-level segmentation network, which learns generic object shapes and can later be transferred to the target dataset.

[Shen CVPR 18]

Webly Supervised Learning

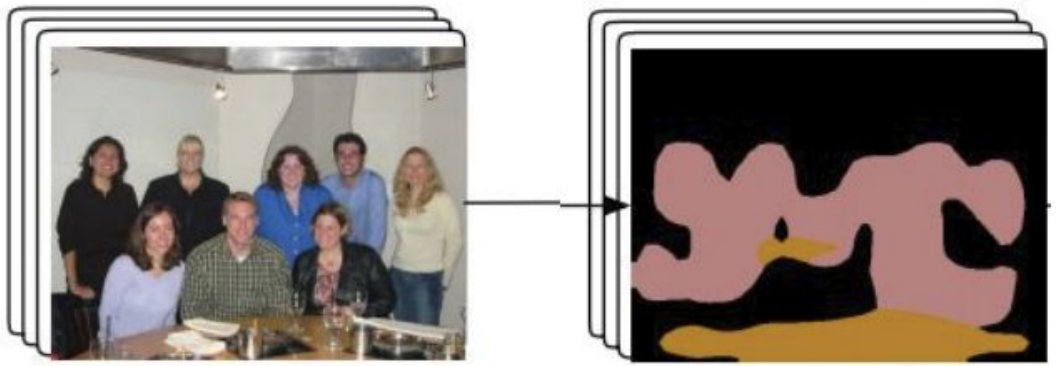
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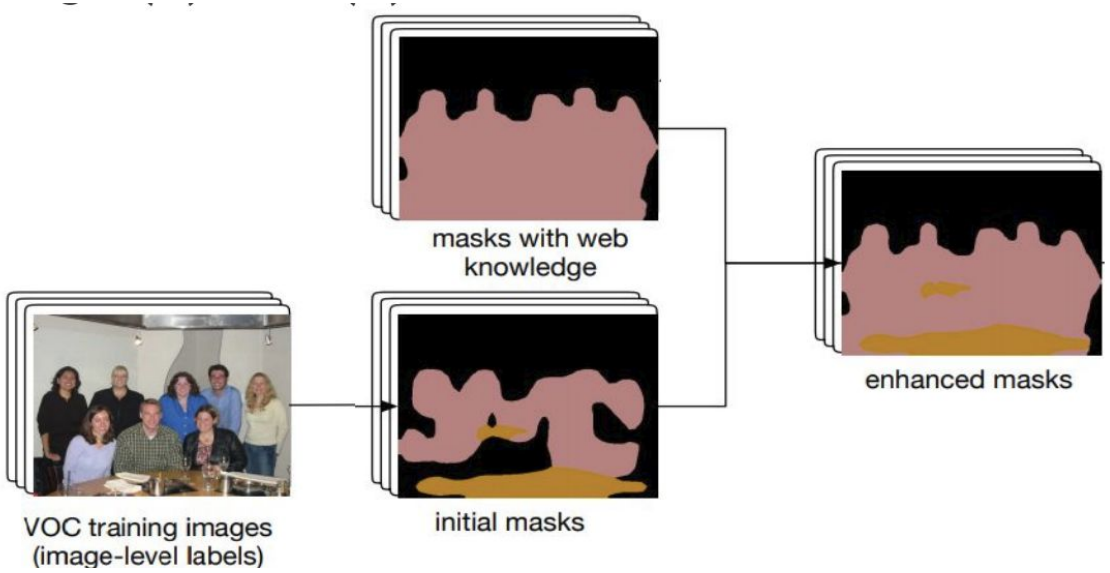
VOC training images (image-level labels)

initial masks

[Shen CVPR 18]

Webly Supervised Learning

- Pipeline
  - 1. Train an image classifier (Model-T) on clean image-level labels.
  - 2. Use Model-T to filter noisy web images.
  - Learn a pixel-level object labeler (Web-SEC) from the filtered web images.
  - 4. Apply Web-SEC to generate pixel labels for the target images.
  - 5. Use Model-T to produce class activation maps (CAMs) on the target images.
  - **6. Fuse the two label sources and train the final segmentation model.**

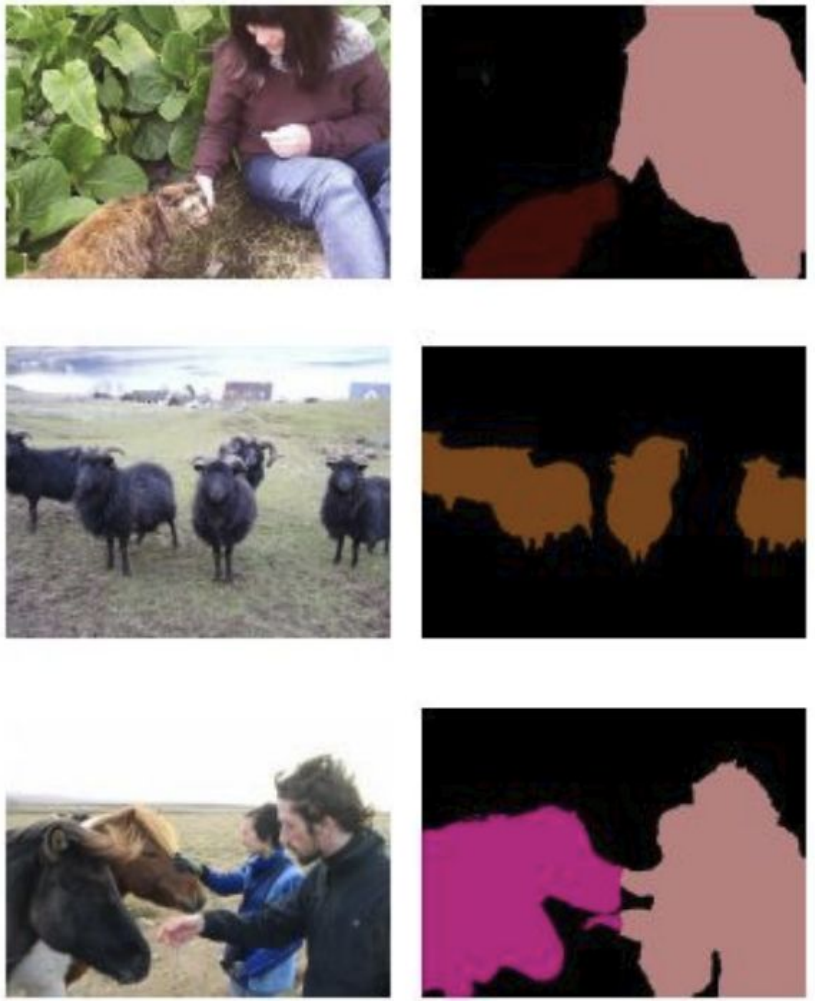


[Shen CVPR 18]

Webly Supervised Learning

Method	val	test	Extra Supervision
Chen *et al.* [2]	**63.7**	**66.4**	Fully supervised
Lin *et al.* [16]	63.1	-	Scribble
Dai *et al.* [6]	62.0	64.6	Bounding box+MCG
Oh *et al.* [21]	55.7	56.7	Bounding box
Bearman *et al.* [1]	46.1	-	Point
Wei *et al.* [29]	55.0	55.7	Supervised saliency
STC [30]	49.8	51.2	Supervised saliency
EM-Adapt [22]	33.8	39.6	-
CCNN [23]	35.3	35.6	-
SEC [13]	50.7	51.7	-
Hong *et al.* [10]	58.1	58.7	-
Ours-VGG16	58.8	60.2	Web
Ours-Res50	**63.0**	**63.9**	Web

Table 1: Comparison with methods using other supervisions.

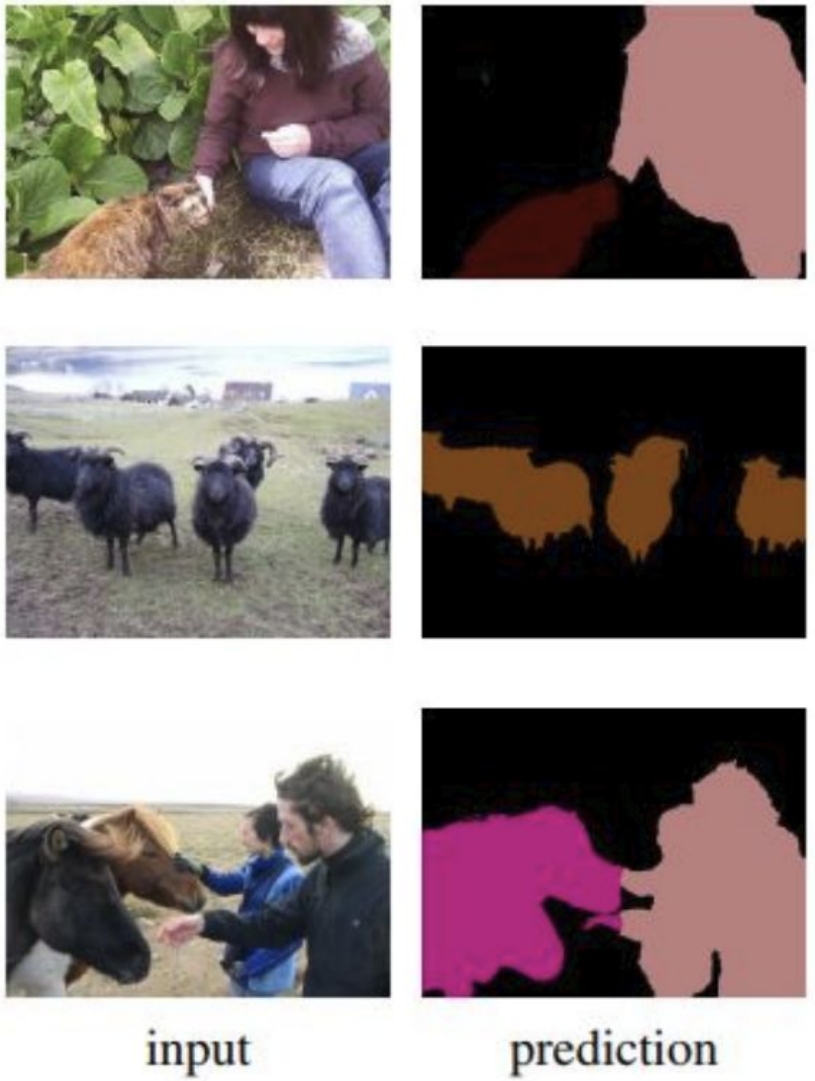


[Shen CVPR 18]

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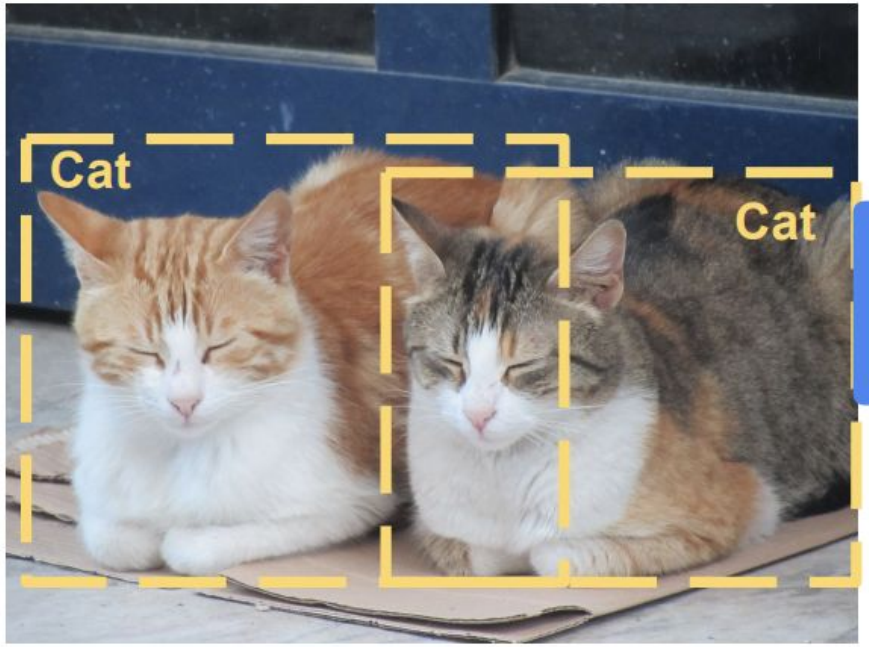
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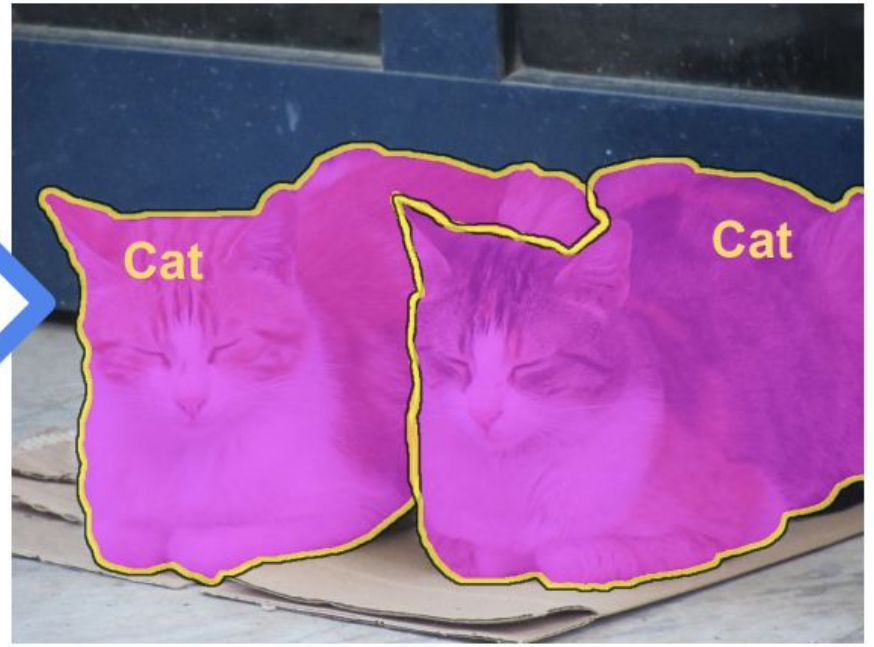
[Shen CVPR 18]

Boxly Supervised Learning

From box-level labels to pixel-level instance labels



Box-level labels
at training time



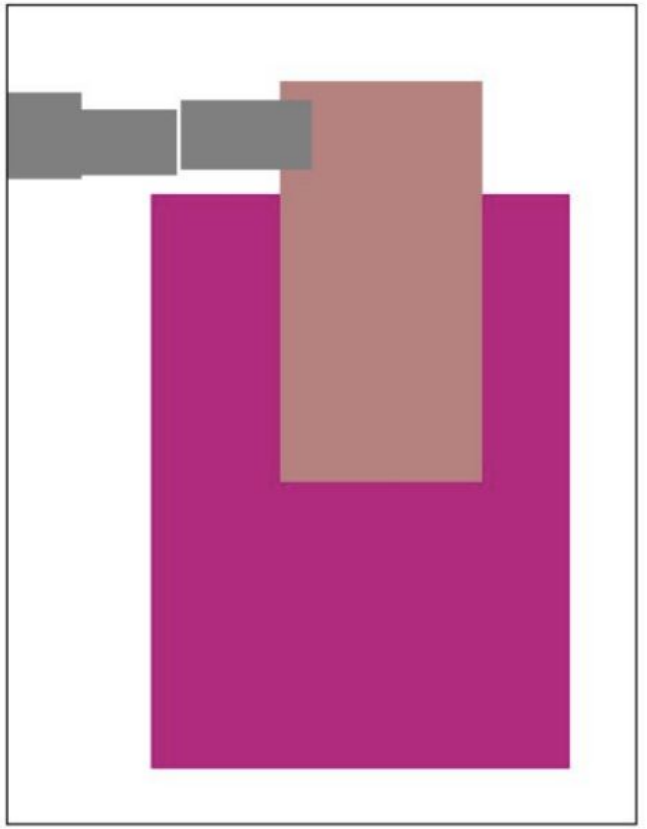
Pixel-level per
instance labels
at test time



What we want:

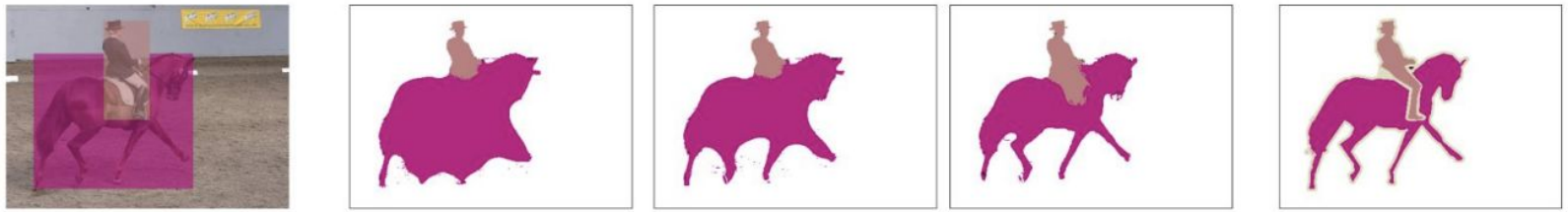


What we have:



[Khoreva CVPR 17]

Convnets are robust to noise. Let us use that !

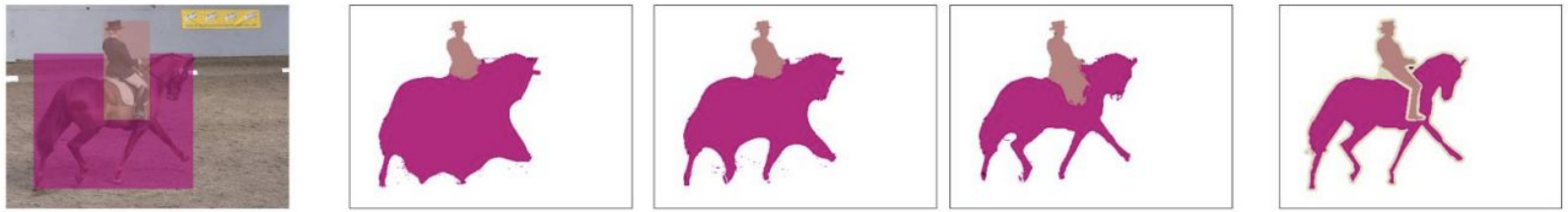


Initially, training is performed using a noisy mask, but the CNN's predictions are then used as pseudo-labels for retraining.

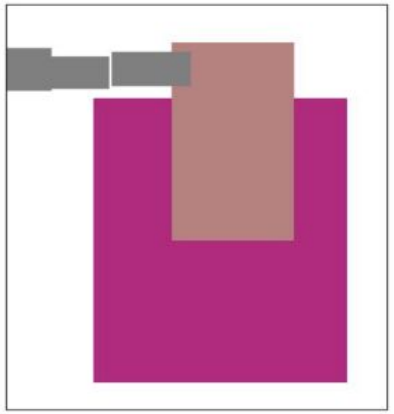
- Segmentation networks are robust to noisy supervision.
- Approximate masks are sufficient for training.
- Iteratively refine pseudo masks during training.
- Better masks lead to better segmentation.

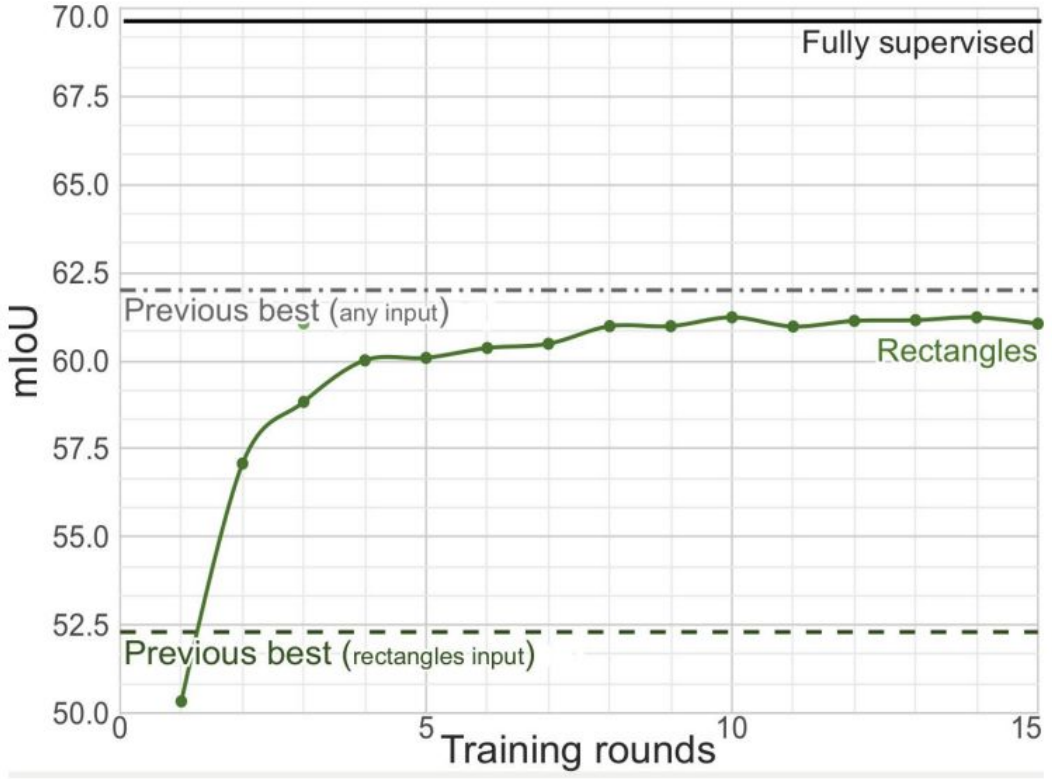
[Khoreva CVPR 17]

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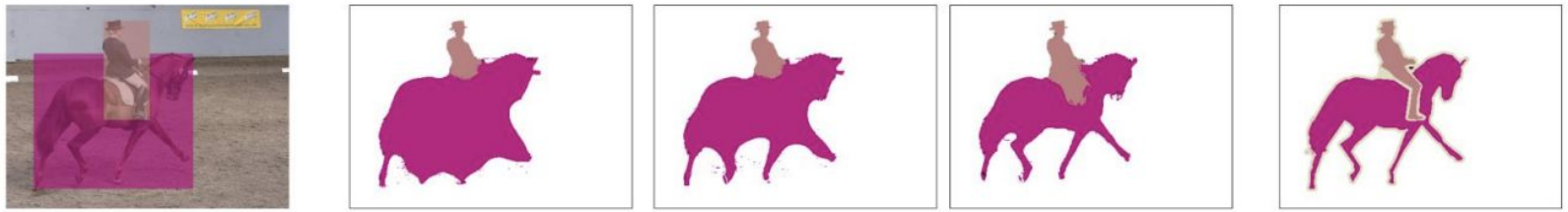
Example input rectangles | Output after 1 training round | After 5 rounds | After 10 rounds | Ground truth



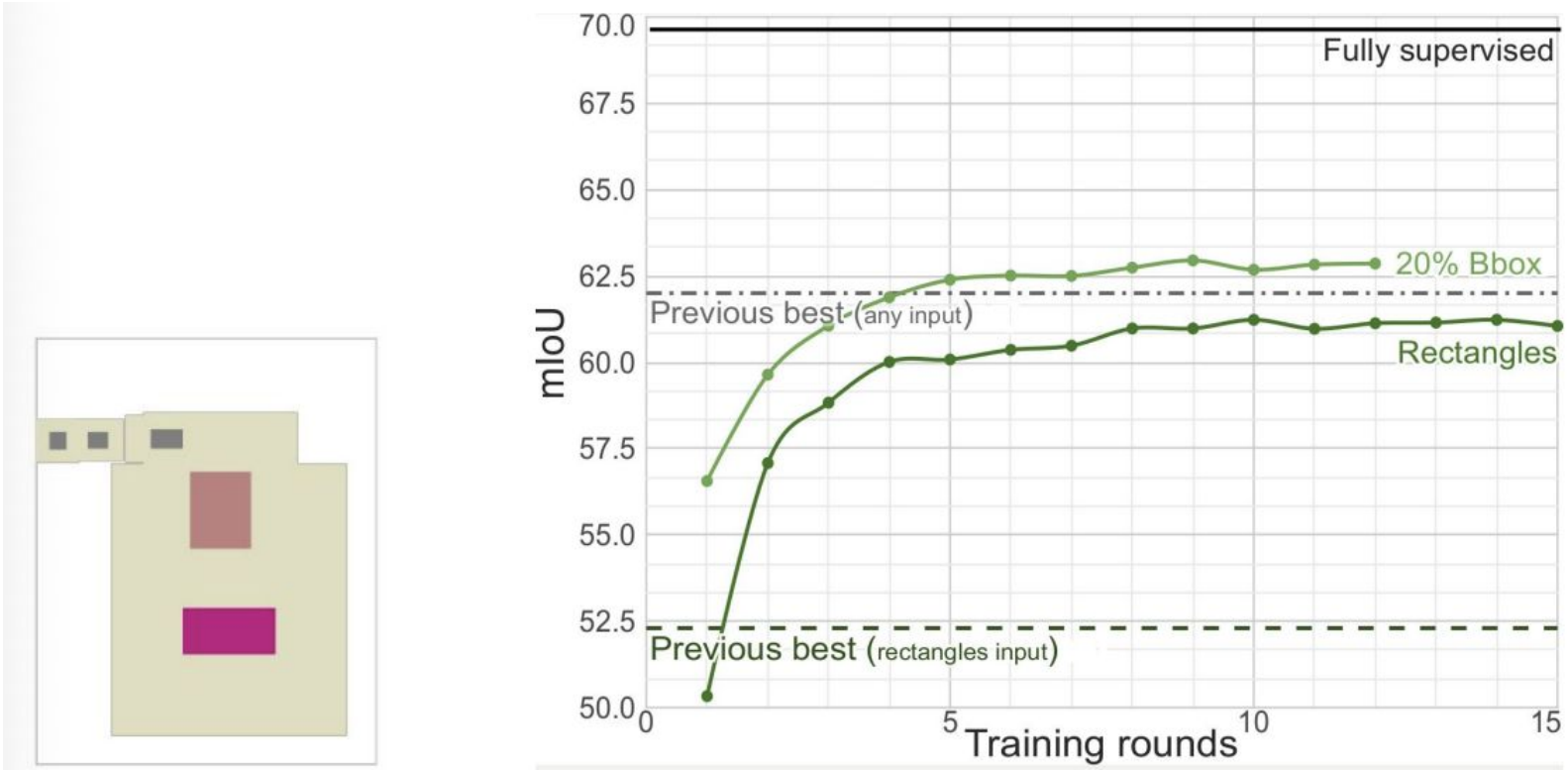


[Khoreva CVPR 17]

Convnets are robust to noise. Let us use that !

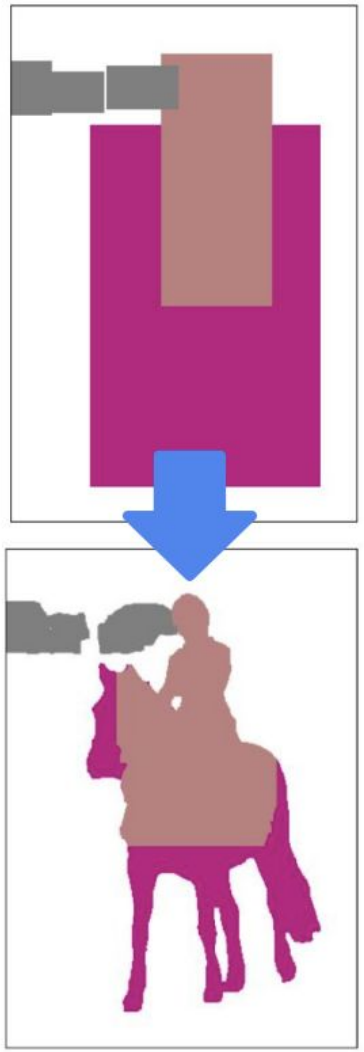


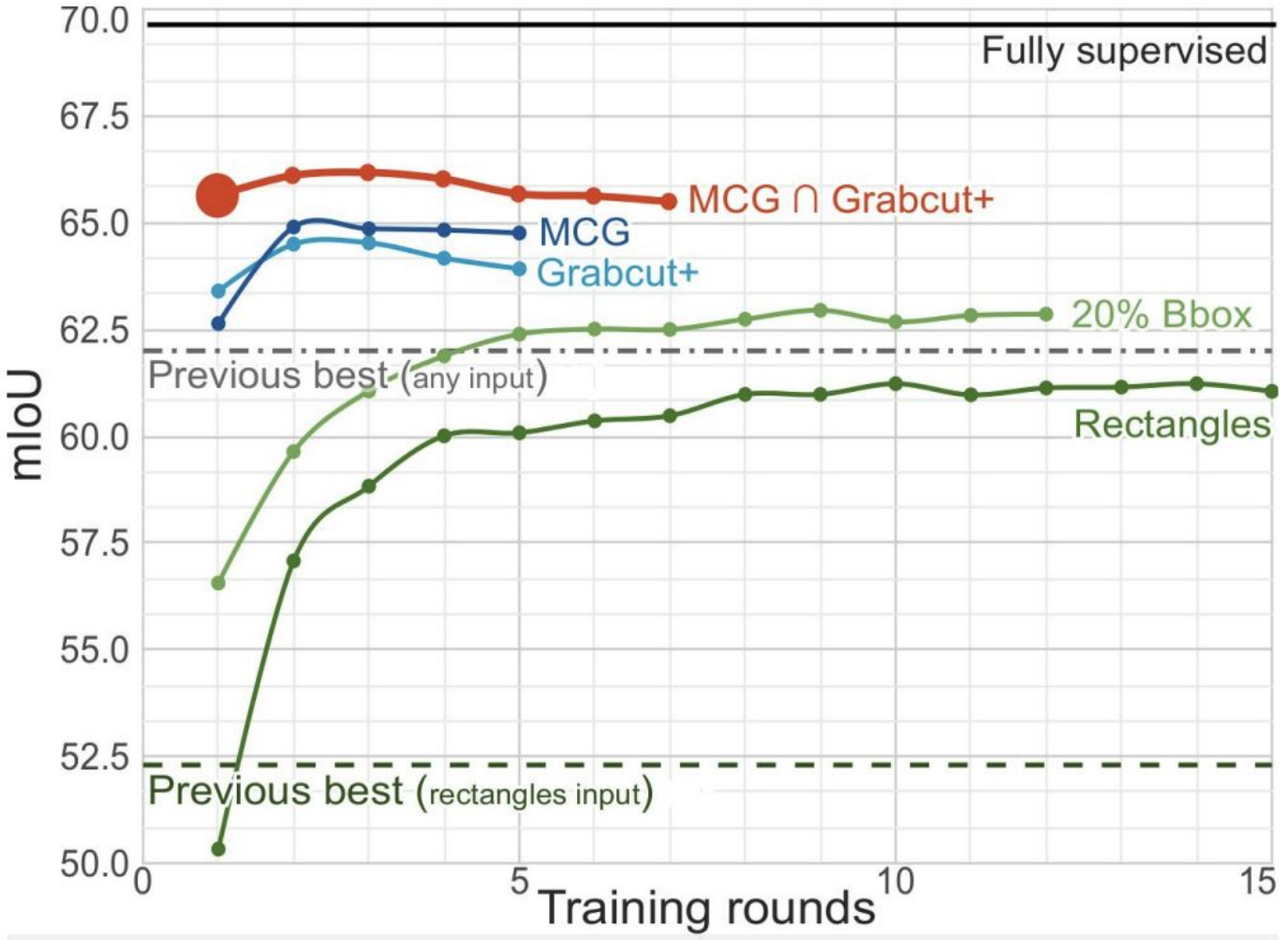
Example input rectangles | Output after 1 training round | After 5 rounds | After 10 rounds | Ground truth



[Khoreva CVPR 17]

Convnets are robust to noise. Let us use that !





[Khoreva CVPR 17]

We reach ~95% of full supervision by using GrabCut++



[Khoreva CVPR 17]

Summary

- Many priors and hints have been explored
- Exploit all available priors, explore new information sources
- Do not underestimate the power of transfer learning (and webly supervision)
- Current techniques are quite effective, and ride the wave of better network architectures

Beyond CAM

CAM activates only the most discriminative regions

CAM boundaries are coarse and inaccurate

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Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

Leveraging Class Distributions in CLIP for Weakly Supervised Semantic Segmentation

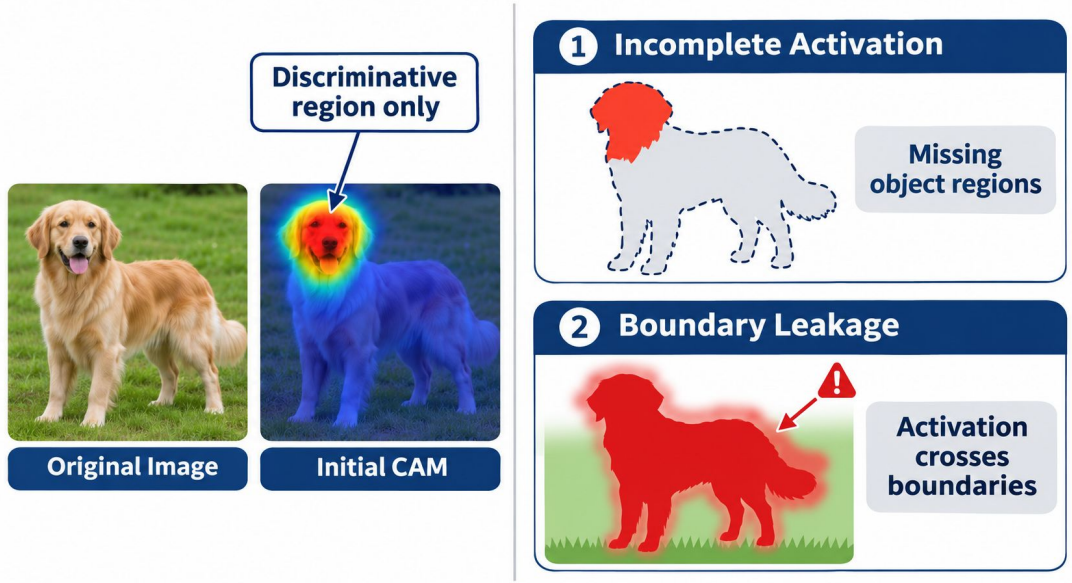
Ziqian Yang[1,2] Xinqiao Zhao[1,2] Xiaolei Wang[1,2] Quan Zhang[1,†] Jimin Xiao[1,†]

[1]Xi'an Jiaotong-Liverpool University [2]University of Liverpool

Yang et.al., Leveraging Class Distributions in CLIP for Weakly Supervised Semantic Segmentation, CVPR 2026

Can CLIP Recover the Whole Object from Image-Level Labels?

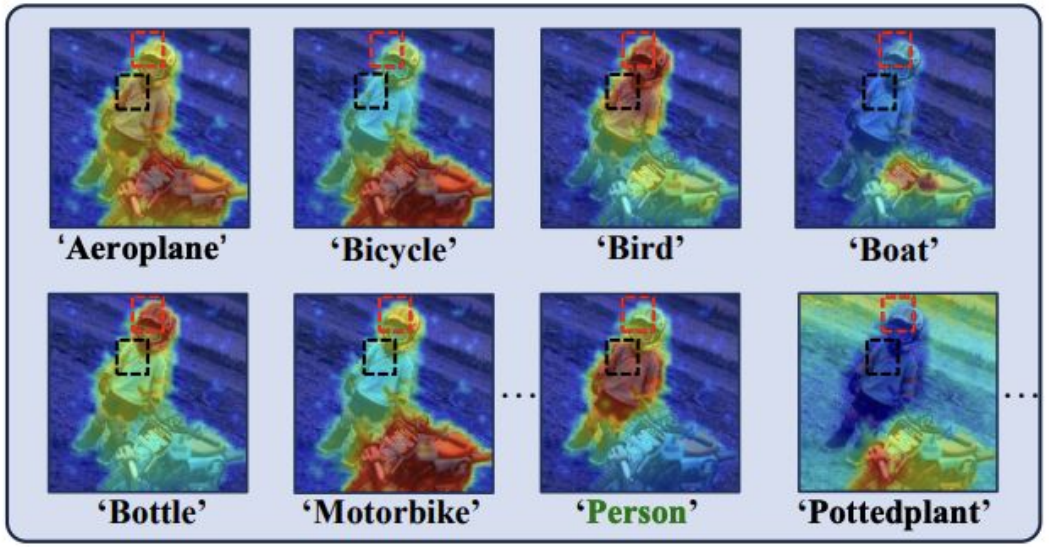
- Problem 1. **Incomplete activation**
  - CAM focuses on the most discriminative part.
  - Other object regions remain inactive.
- Problem 2. **Inaccurate boundaries**
  - Affinity propagation may spread activation.
  - Foreground activation can cross class boundaries.



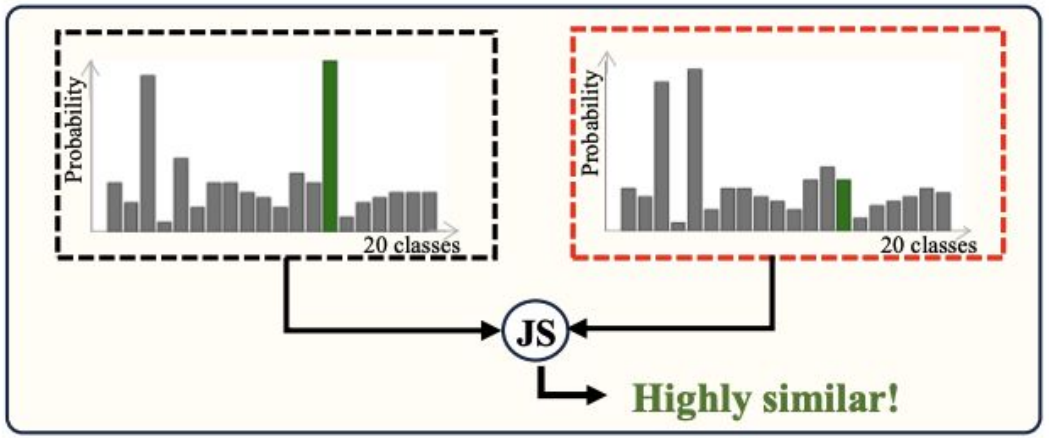
How can we expand CAMs without losing object boundaries?

Target Score Is Not Enough

- Conventional affinity
  - Uses feature or attention similarity between patches.
  - Similar appearance does not always imply the same class.
  - Incorrect affinity causes under-activation or boundary leakage.
- CD-CLIP observation
  - Every **patch** has scores **over multiple classes**.
  - Patches of the same semantic class show **similar class distributions**.
  - The full distribution can be more informative than one target-class score.



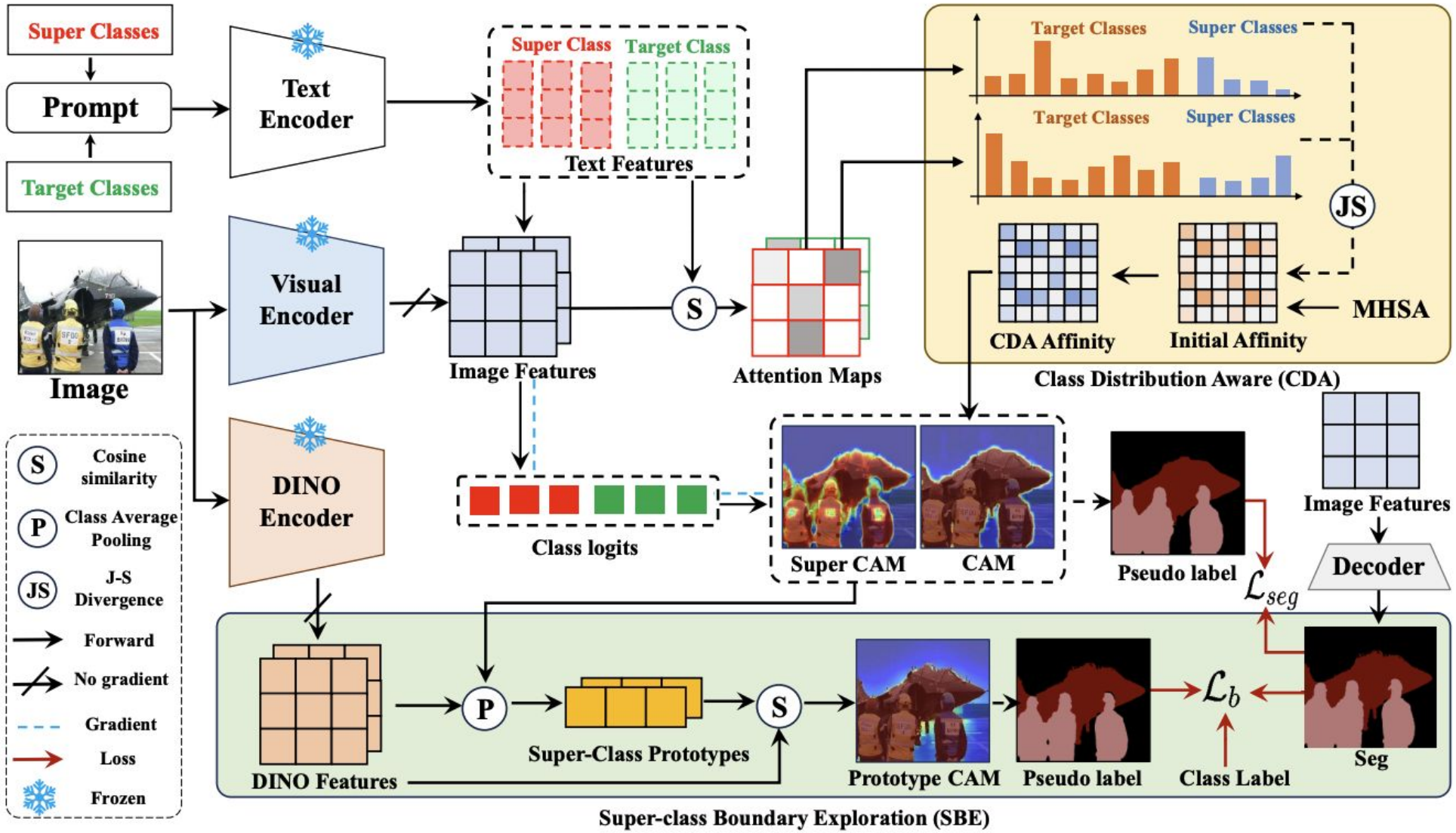
(a) CAMs refined by MHSA Affinity



(b) Class Distributions

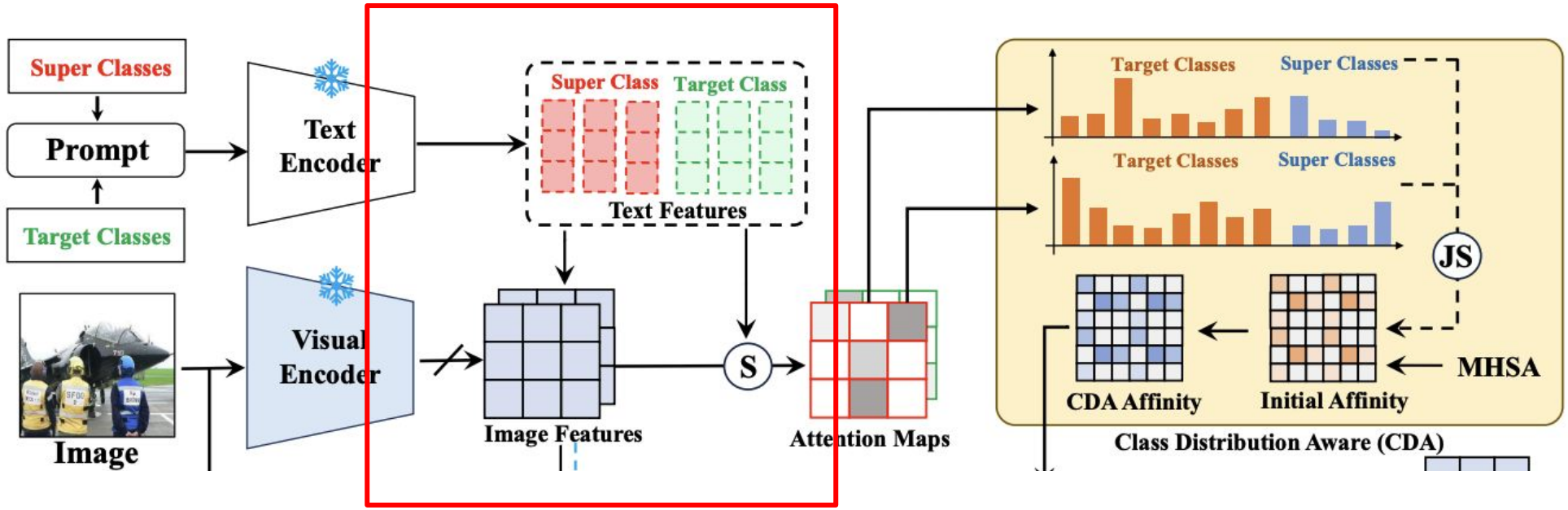
A patch is characterized not only by its target-class score, but by its distribution over all classes.

An overview of our framework CD-CLIP



How can we determine whether two patches belong to the same semantic region?

Each Patch Has a Class Distribution

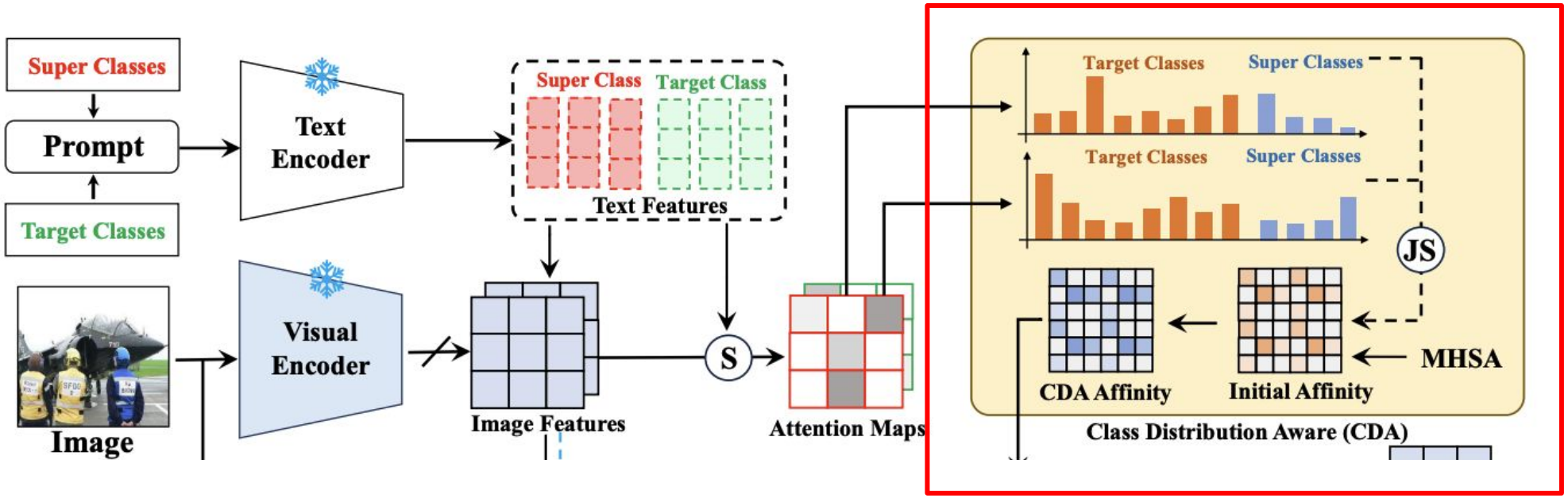


A patch is described by its distribution over classes, not only by one target-class score.

$$\mathbf{S_a} = \text{Min-max}(\text{CosSim}(\mathbf{F}, \mathbf{T}_{all})),$$

- Target classes provide specific semantics.
- Super-classes provide broader semantic relationships.
- Each patch is represented by its scores over all classes.

From Class Distributions to Patch Affinity

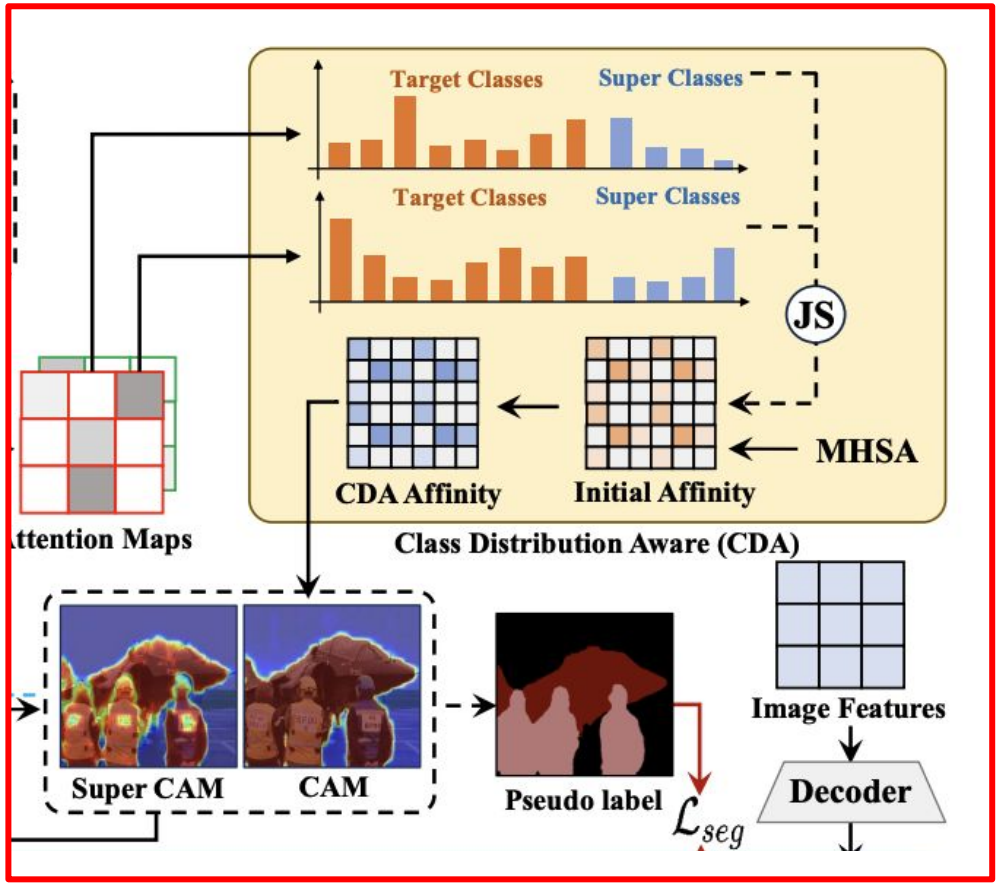


How are similar class distributions used to complete the CAM?

From Class Distributions to Patch Affinity

- 1. Compare two **patch distributions**
  - Small JS divergence: similar class distributions
  - Large JS divergence: different class distributions
- 2. **Correct** the original **affinity**
  - Distribution similarity filters unreliable relationships in the original affinity.
- 3. **Propagate** CAM activation
  - The head is strongly activated in the initial CAM.
  - The body has a similar class distribution.
  - CDA propagates activation from the head to the body.

CDA connects semantically similar patches and propagates activation to missing object regions.



$$\mathbf{S_d} = \frac{(1 - \mathbf{D_{JS}}(\mathbf{Softmax}(\mathbf{S_a}) || \mathbf{Softmax}(\mathbf{S_a})^{\mathrm{T}}))}{\tau},$$

$$\mathbf{A}_d = \mathbf{Softmax}(\mathbf{S_d}) \cdot \mathbf{A}_{init},$$

$$\mathbf{M}^c_{re} = \mathbf{M}^c_{init} \cdot (\mathbf{A}_d)^t.$$

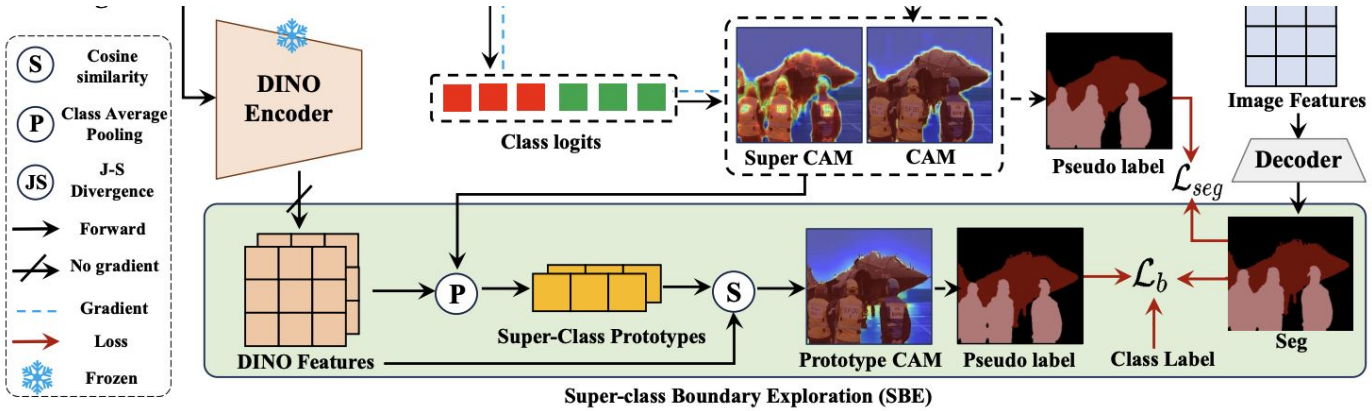
Complete CAMs Can Cross Boundaries

- CDA expands activation—but where should it stop?
  - 1. Construct a **super-class prototype**
    - Use the **CLIP** super-class **CAM** as a semantic seed and average the corresponding DINO features.

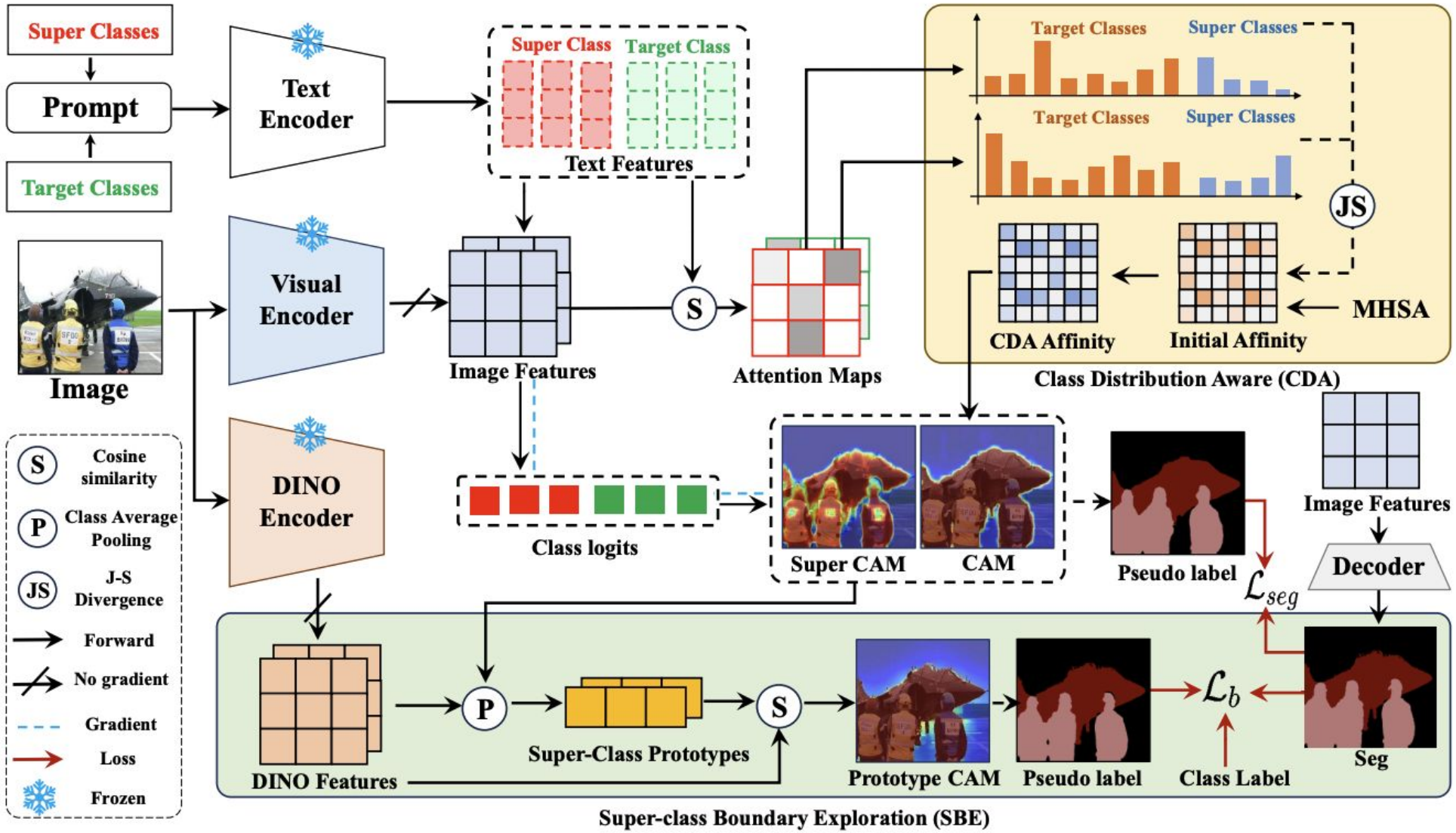
$$\mathbf{f}_s^n = \mathbf{CAP}\left(\mathbf{M}_s^n \odot \mathbf{F}_d\right)$$

  - 2. Generate a **prototype CAM**
    - Compare the prototype with every **DINO** patch to recover a complete and structurally coherent super-class region.

$$\mathbf{M}_p^n = \mathbf{ReLU}(\mathbf{CosSim}(\mathbf{f}_s^n, \mathbf{F}_d))$$



CLIP gives the prototype a semantic identity, and DINO expands it into a coherent region



Training objective

- Boundary-enhanced loss
  - Apply boundary loss only when there are two or more distinct superclasses.

$$\mathcal{L} = \mathcal{L}_{seg} + \lambda_1 \mathcal{L}_b.$$

$$\mathcal{L}_b = \begin{cases} \mathbf{Dice}(\mathbf{P}, \mathbf{Y}_s) & \text{if} \quad |\mathbf{C}_\mathrm{p}| \geq 2, \\ 0, \text{otherwise.} \end{cases}$$

Does Each Component Help?

- CDA significantly improves the CAM mIoU from 70.1 to 80.3.
- SBE provides a greater improvement to the final segmentation performance than the CAM itself.
- This is because SBE contributes to boundary learning within the decoder, rather than to activation expansion.

Table 3. Ablation study of two modules in our method, 'M' denotes the mIoU (%) of CAM performance. 'Seg.' denotes the mIoU (%) of segmentation performance.

#	CAA	RFM	CDA	SBE	M	Seg.
0	No CAM refinement				70.1	68.8
1	✓				73.2	72.8
2		✓			77.4	76.1
3			✓		80.3	80.2
4	✓		✓		76.8	76.4
5		✓	✓		80.1	80.2
6			✓	✓	80.8	81.6

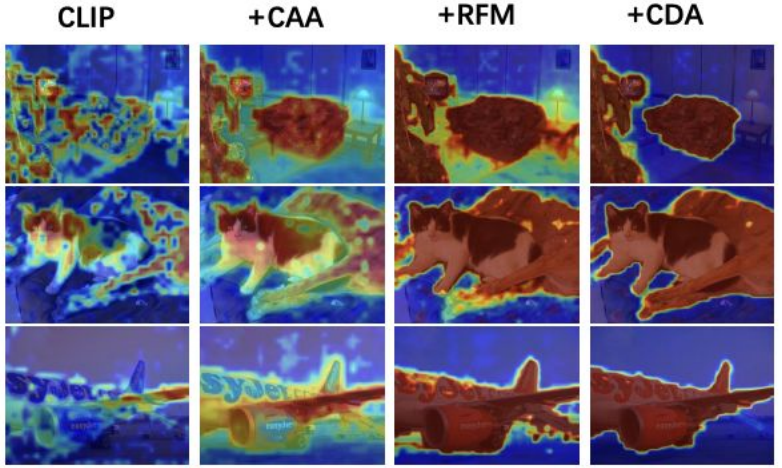
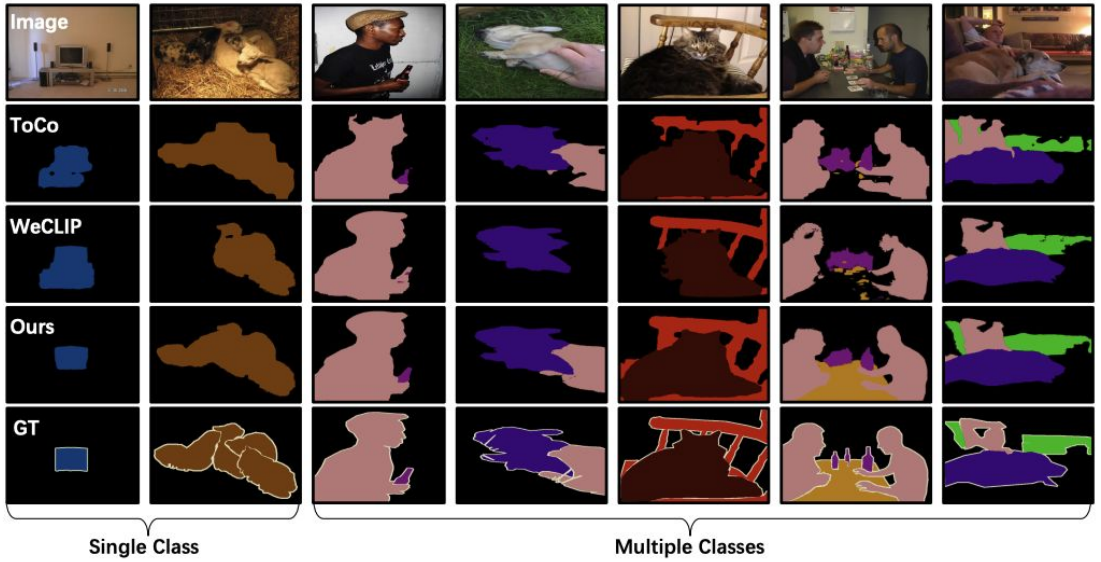


Figure 4. Visualization comparison of CAMs between other modules and our CDA method. The experimental dataset is PASCAL VOC 2012 train split.

Boundary supervision is most meaningful when different semantic super-classes coexist.

Final Results and Takeaway

- CD-CLIP addresses two complementary problems:
  - 1. CDA: CAM completeness
    - Class distributions → better patch affinity → complete CAM
  - 2. SBE: Boundary accuracy
    - DINO structure → super-class prototypes → boundary supervision



Better WSSS requires both knowing which patches belong together and knowing where their relationship should stop.

Can a segmentation model learn new classes from the web without storing its original training images?

Learning from the Web: Language Drives Weakly-Supervised Incremental Learning for Semantic Segmentation

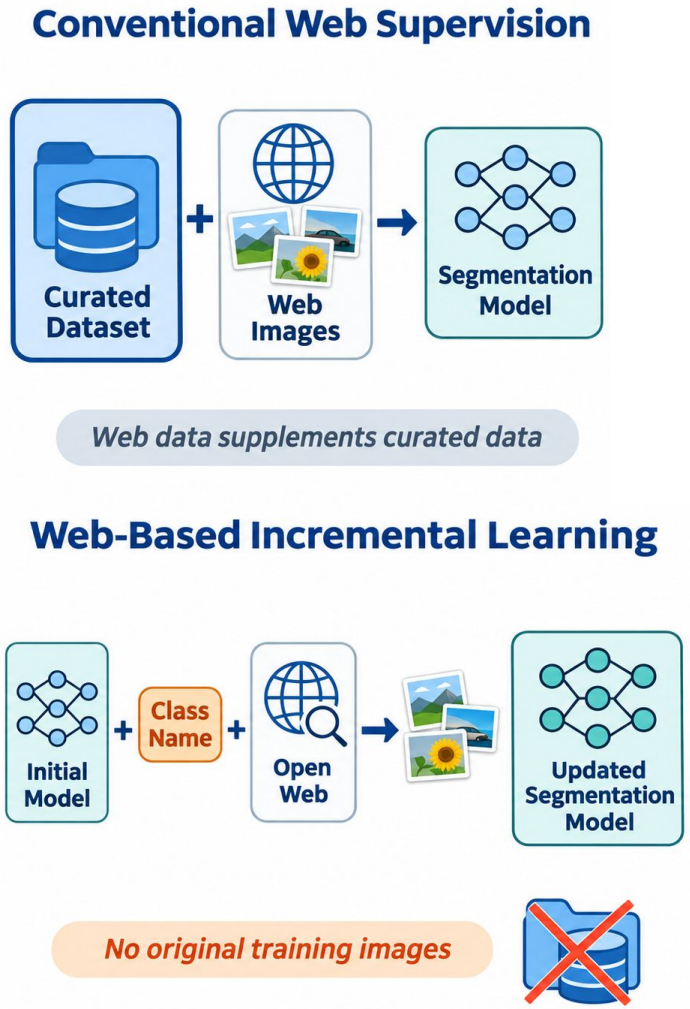
Chang Liu[1], Giulia Rizzoli[2], Pietro Zanuttigh[2], Fu Li[1], and Yi Niu[1]*

[1] School of Artificial Intelligence, Xidian University, China
[2] Department of Information Engineering, University of Padova, Italy

Liu et.al., Learning from the Web: Language Drives Weakly-Supervised Incremental Learning for Semantic Segmentation, ECCV 2024

Can the Web Replace the Training Dataset?

- Can a segmentation model learn **new classes** from the **web without storing** its original training images?
- Conventional web supervision
  - Web images supplement a curated training dataset.
  - Target classes are fixed.
  - Previous training images remain available.
- New scenario
  - Only the initial model is retained.
  - **Original training** images are **no** longer **accessible**.
  - **New** and **old** classes must be learned **from web** images.

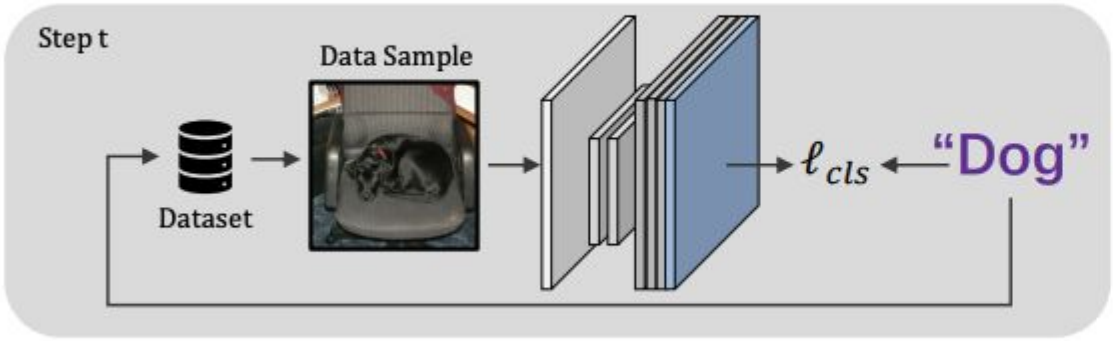


Can the Web Replace the Training Dataset?

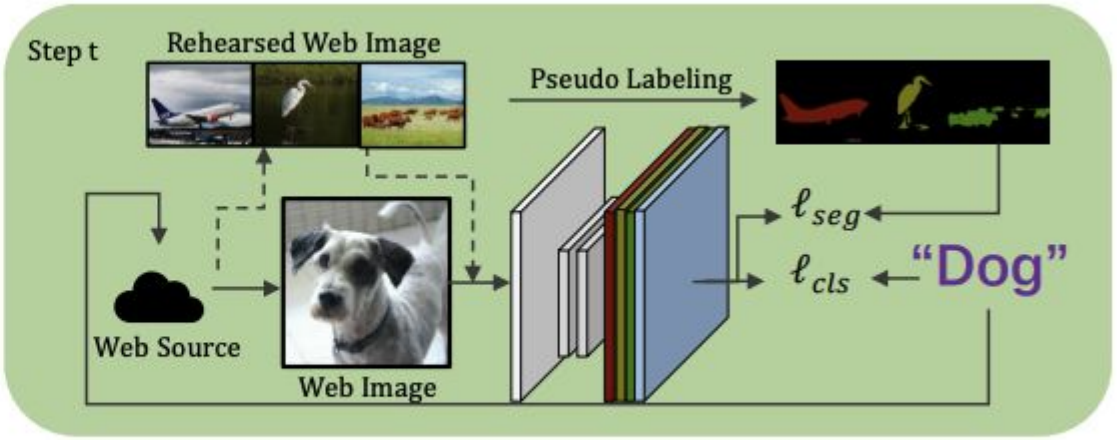
- Three challenges
  - 1. Web images have a **different distribution**.
  - 2. Search queries provide **noisy** single-class labels.
  - 3. Learning new classes causes **forgetting** of old classes.

The web must provide both new knowledge and memory of old knowledge.

From Curated Data to Open-Web Learning



(a) WILSS



(b) Ours

Conventional WILSS

- A new class is introduced at step t.
- Curated dataset images are available.
- Only image-level labels are given for new classes.
- The localizer generates pseudo segmentation labels.

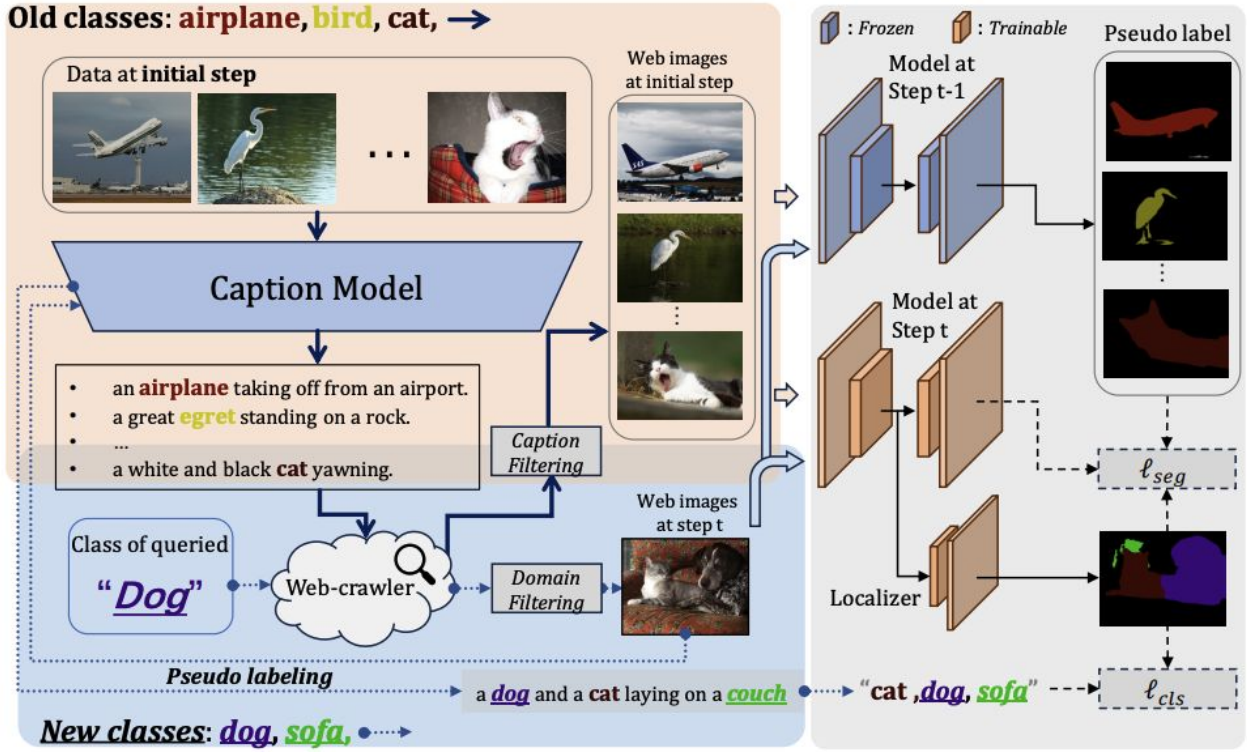
Proposed Setting

- New images are downloaded from the **web**.
- Old-class web images are also retrieved for rehearsal.
- **Caption** models provide richer image-level supervision.
- **No original** image is **stored** after the initial step.

Web images replace curated incremental data, while captions replace reliable manual image labels.

Language-Driven Web Learning

- ① **Retrieve** web images
  - Query the web using new class names.
  - Retrieve rehearsal images using stored captions.
- ② **Filter noisy** web images
  - Domain filtering removes out-of-distribution images.
  - Caption filtering removes semantically irrelevant images.
- ③ **Generate supervision**
  - Captions are converted into multi-label supervision.
  - The previous model produces pixel-level pseudo-labels.
- ④ **Update** the segmentation **model**
  - Learn new classes from web images.
  - Rehearse old classes without storing original images.

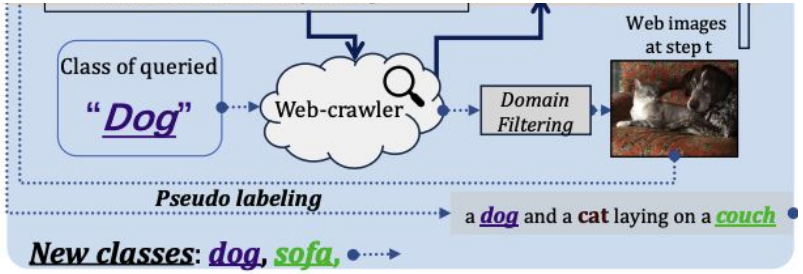


Web retrieval → Filtering → Caption labeling →Pseudo labeling → Incremental training

Language is used for querying, labeling, filtering, and rehearsal.

Learning New Classes from the Web

- 1. Select Dataset-Like Web Images
  - Query the web using a new class name.
  - Remove web images with a **large domain gap**.
  - **Keep** images that **resemble** the original dataset.
- 2. Recover Multi-Label Supervision
  - A search query provides only one label.
  - A caption describes all visible objects.
  - **Caption** words are **converted** into **image-level labels**.



$$(p_{ds}, p_{web}) = (p(\mathbf{x} \in \mathcal{X}), p(\mathbf{x} \in \mathcal{X}^{web})) = M_D(|\mathcal{F}(\mathbf{x})|)$$

$$p_{ds}/p_{web} > 1.$$

The class name retrieves images; the caption determines their labels.

The Web as External Memory

- Can Captions Replace Stored Training Images?
- **Store Captions**, Not Images
  - Original training images are unavailable after the initial step.
  - The method stores their captions instead.
  - Stored captions are used to retrieve rehearsal images from the web.

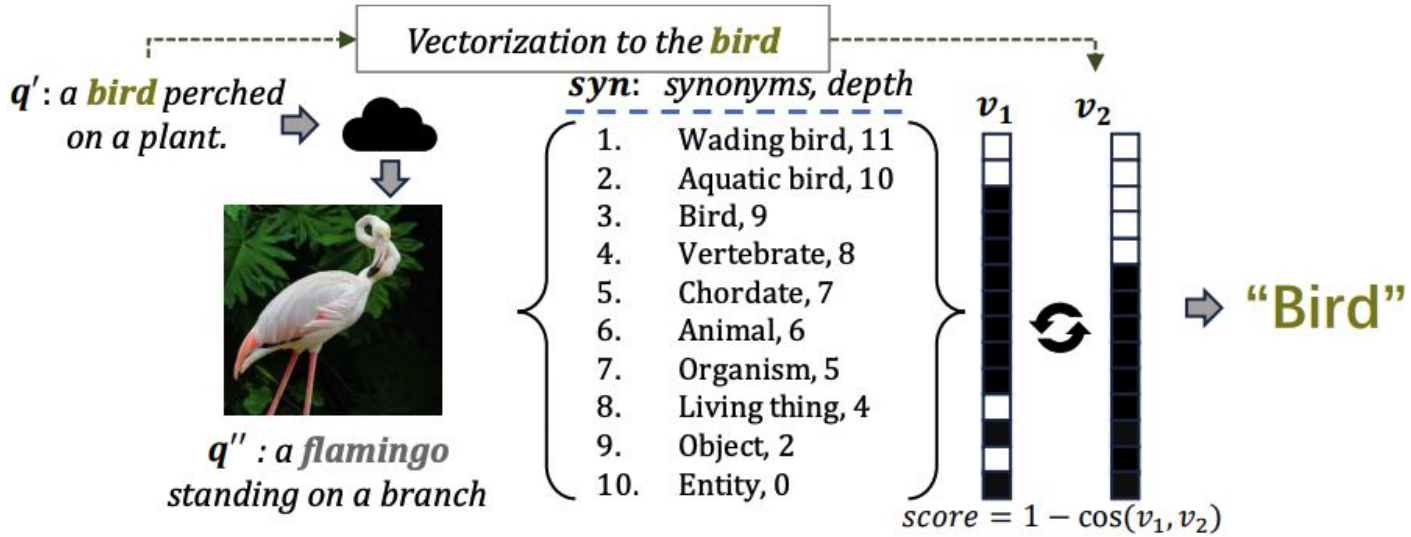
$$\mathcal{X}_r^{web} = \{\mathbf{x} = \mathcal{D}^{web}(q') \,|\, q' = M_{CAP}(\mathbf{x}) : \mathbf{x} \in \mathcal{X}\}$$

Old image → Stored caption q' → Web search → Rehearsal image

q': caption of an old training image

The Web as External Memory

- Verify the Retrieved Image
  - 1. Generate a new caption q'' for the retrieved image.
  - 2. Compare the nouns in q' and q''.
  - 3. Keep the image if they belong to a similar semantic family.



Captions serve as retrieval keys, and the web serves as external memory.

Do Filtering and Captions Actually Help?

Table 4: Ablation study on the web learning strategies: Pascal-VOC Overlapped.

Learning from Web		10-10			15-5		
Domain filtering	Caption Labeling	1-10	11-20	All	1-15	16-20	All
		72.0	49.7	62.1	74.2	43.8	67.8
✓		72.0	51.9	63.1	73.7	46.3	68.0
	✓	72.6	52.1	63.6	73.8	**46.7**	68.2
✓	✓	**73.8**	**53.9**	**65.0**	**74.4**	45.9	**68.4**

Table 5: Ablation study on the rehearsal strategies on Pascal-VOC Overlapped.

Rehearsal Strategy		10-10			15-5		
Caption query	Caption Filtering	1-10	11-20	All	1-15	16-20	All
		71.6	53.1	63.6	76.0	47.4	69.8
✓		72.2	55.1	64.9	77.6	**53.9**	72.6
	✓	71.5	52.9	63.5	**78.8**	47.8	71.9
✓	✓	**73.0**	**55.2**	**65.3**	78.7	53.7	**73.4**

Learning New Classes

- Domain filtering reduces the web-to-dataset gap.
- Caption labeling corrects incomplete single-class supervision.
- The two components are complementary.
- Combined use improves the 10–10 result by 2.9 points.

Web rehearsal ablation

- Caption queries retrieve semantically richer images than class-name queries.
- Caption filtering removes irrelevant retrievals.
- Querying and filtering work best together.
- The combined method improves 15–5 performance by 3.6 points.

Homogeneous Supervision
- Every training image uses the same weak label type.
- Tags are cheap but provide no localization.
- Boxes are informative but expensive.
- Points provide an intermediate level of supervision.

Web Images Can Provide Both Learning and Rehearsal

- The proposed method improves the challenging 10–1 result substantially.
- Caption-based rehearsal is particularly valuable in multi-step learning.

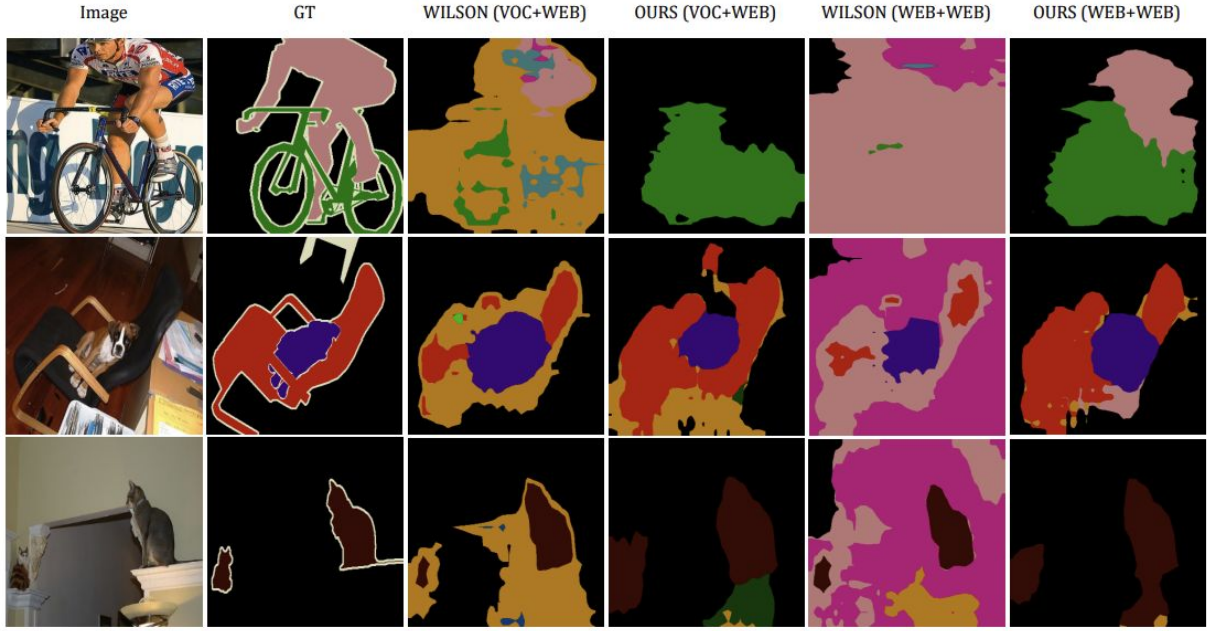


Fig. 4: Qualitative results in the overlapped 10-1 setting.

Table 3: Comparison on single-step multi-class settings on PASCAL-VOC.

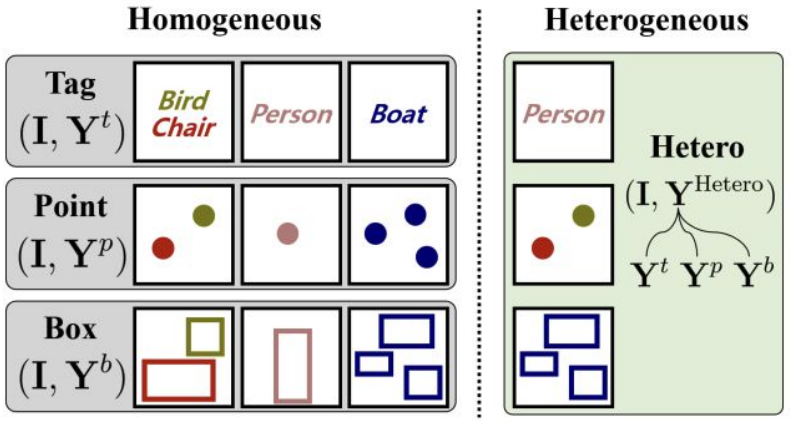
				15-1			10-1		
Method	**Train**	**Memory**	**M**	1-15	16-20	All	1-10	11-20	All
WILSON [7]	VOC	-	-	0.0	2.3	0.6	0.0	0.2	0.1
RaSP [37]	VOC	-	-	17.7	0.9	13.2	2.0	0.7	1.3
RaSP [37]	VOC	VOC	100	63.3	28.3	56.0	38.9	30.9	36.9
WILSON [7]	VOC	ImageNet	100	**75.7**	32.9	65.9	**66.8**	34.9	52.3
RaSP [37]	VOC	ImageNet	100	**75.7**	35.2	66.6	**66.8**	39.1	**54.4**
WILSON	VOC	WEB	100	73.9	29.9	63.6	63.0	35.6	49.9
Ours	VOC	WEB	100	73.6	36.2	65.1	63.9	39.2	52.0
WILSON	WEB	WEB	100	74.4	41.6	67.2	61.0	28.3	42.9
Ours	WEB	WEB	100	74.5	**41.9**	**67.4**	63.7	**42.4**	**54.4**

The web is not only a source of new data. It can serve as an external semantic memory.

Under a fixed annotation budget, should every image receive the same type of weak label?

Must Every Image Use the Same Weak Label?

- Homogeneous Supervision
  - Every training image uses the same weak label type.
  - Tags are cheap but provide no localization.
  - Boxes are informative but expensive.
  - Points provide an intermediate level of supervision.
- Heterogeneous Supervision
  - Different images may use **different weak label** types.
  - **Cheap** labels provide **broad** data coverage.
  - **Strong** labels provide **accurate** localization.
  - A **single model** learns from **all label types**.

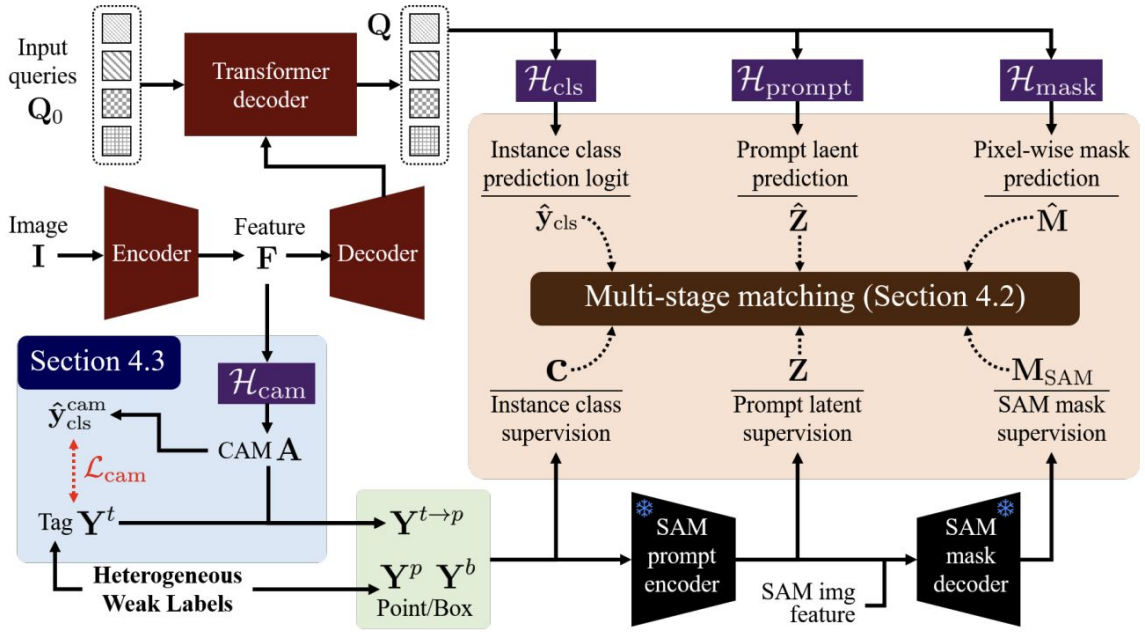


$$\mathbf{Y}_i^{\text{Hetero}} = \mathbf{Y}_i^t \text{ or } \mathbf{Y}_i^p \text{ or } \mathbf{Y}_i^b.$$

Class matching identifies the semantic category.
Prompt matching identifies the approximate instance location.
Mask matching transfers SAM's segmentation knowledge.
Hungarian matching assigns predictions to weakly labeled instances.

WISH: A Common Language for Weak Labels

- SAM's prompt space serves as a common representation for heterogeneous supervision.



Predict prompt latent from object queries

- Q: Mask2Former object queries
- H_prompt: prompt prediction head
- hat(z): predicted prompt latent

$$\hat{\mathbf{Z}} = \mathcal{H}_{\text{prompt}}(\mathbf{Q}),$$

Weak labels to prompt latent

- Point prompt → point latent
- Box prompt → box latent
- Tag → converted point prompt

- SAM maps points and boxes into a shared prompt space.
- WISH predicts the corresponding prompt latent from each object query.
- Weak labels with different formats become comparable in one space.
- The model learns what instance each query should represent.

How Is Each Prediction Matched to a Weak Label?

- 1. Class Matching
  - Does query i predict the class of weak label j?

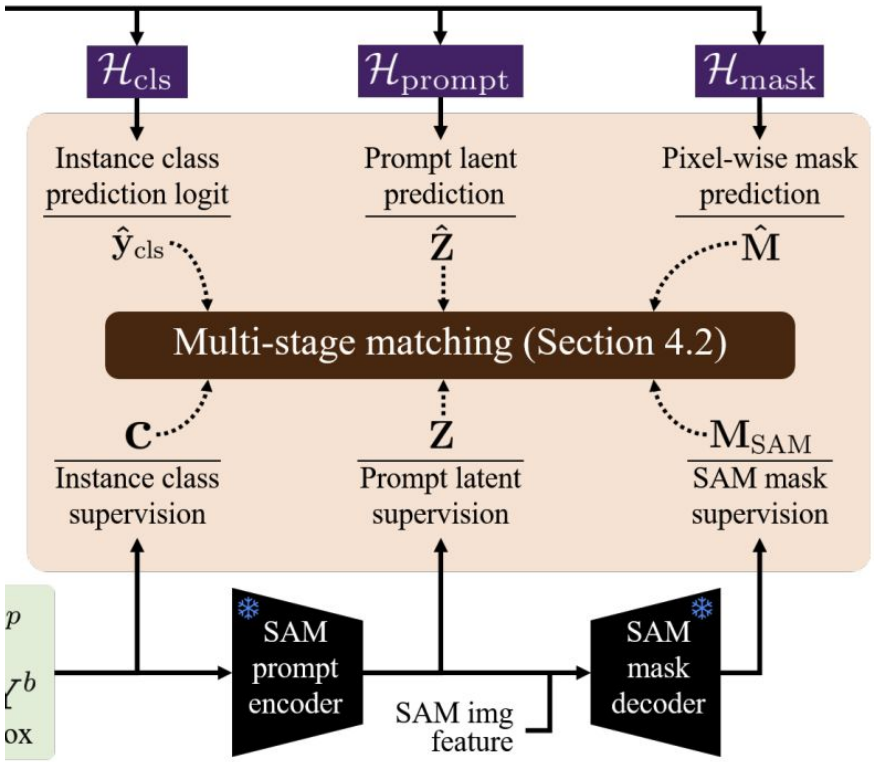
$$\text{Cost}_{\text{cls}}^{i,j} = \hat{\mathbf{y}}_{\text{cls}}^{i}(\mathbf{c}^{j})$$

- 2. Prompt Matching
  - Does the predicted query represent a similar point or box prompt?

$$\text{Cost}_{\text{prompt}}^{i,j} = \text{KLD}(\hat{\mathbf{Z}}^{i}, \mathbf{Z}^{j})$$

- 3. Mask Matching
  - Select the SAM candidate mask closest to the predicted instance mask.

$$\text{Cost}_{\text{mask}}^{i,j} = \min_{n\in\{1,2,3\}} d(\hat{\mathbf{M}}^{i}, \mathbf{M}_{\text{SAM}}^{j,n})$$



$$\text{Cost}^{i,j} = \alpha\text{Cost}_{\text{cls}}^{i,j} + \beta\text{Cost}_{\text{prompt}}^{i,j} + \gamma\text{Cost}_{\text{mask}}^{i,j}$$

How Is Each Prediction Matched to a Weak Label?

- Class matching identifies the **semantic category**.
- Prompt matching identifies the approximate instance **location**.
- Mask matching transfers **SAM's** segmentation **knowledge**.
- Hungarian matching assigns predictions to weakly labeled instances.

WISH matches an instance by asking what it is, where it is, and which SAM mask fits it best.

How Can a Class Tag Become an Instance Prompt?

- 1. Generate Class Activation Maps
  - **CAM** provides rough **class-specific localization**.
  - **Local maxima** are extracted from each class activation map.

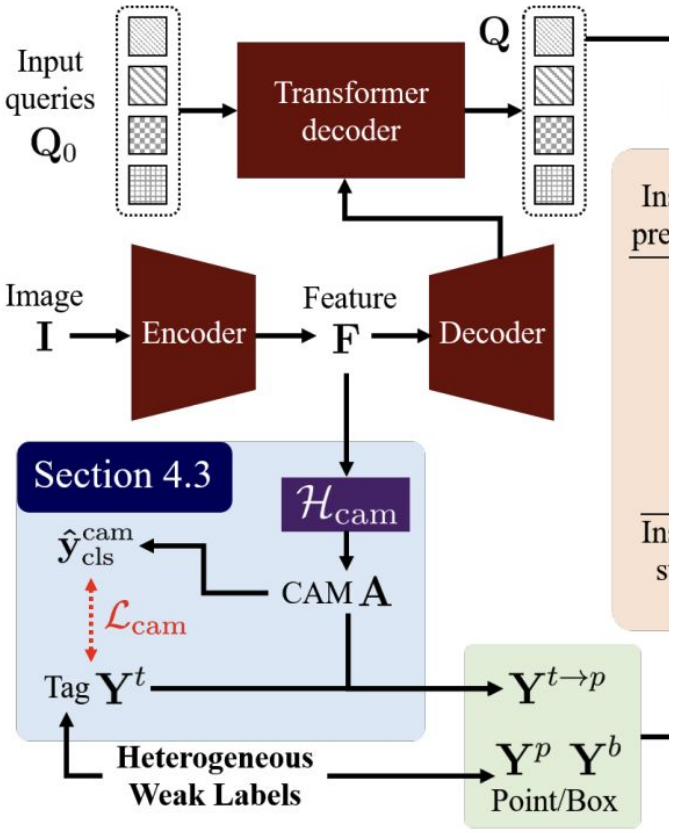
$$\mathbf{A} = \mathcal{H}_{\text{cam}}(\mathbf{F})$$

- 2. Convert CAM Peaks into Point Prompts
  - Each selected **CAM peak** becomes a positive point prompt.
  - **SAM** generates an **instance mask** from each point.
  - Overlapping masks are grouped to identify individual instances.

$$\mathbf{Y}_i^{t \to p} = \{(\mathbf{X}_i^1, \mathbf{c}_i^1), \ldots, (\mathbf{X}_i^{k_i'}, \mathbf{c}_i^{k_i'})\}$$

- 3. Improve CAMs with Model Predictions
  - Each selected CAM peak becomes a positive point prompt.
  - SAM generates an instance mask from each point.
  - Overlapping masks are grouped to identify individual instances.

$$\mathcal{L}_{\text{self}} = \mathcal{L}_{\text{bce}}(\mathbf{A}, \hat{\mathbf{M}}_c)$$



CAM supplies missing spatial cues, and SAM turns those cues into instance-level masks.

Which Labels Should We Buy?

- Tags are cheap, so many images can be labeled.
- Boxes are expensive, so fewer images can be labeled.
- Strong supervision and broad data coverage must be balanced.
- WISH searches for effective heterogeneous combinations.

Table 3. Results of WISH framework under heterogeneous settings on PASCAL VOC val set. The top row provides the results of tag-point while the bottom row shows the results of tag-box.

N_t	10,582	7,936	5,290	2,644	0
N_p	0	441	882	1,323	1,764
AP	46.0	46.9	**47.3**	47.2	44.7
N_t	10,582	7,936	5,290	2,644	0
N_b	0	220	441	661	882
AP	46.0	47.2	**48.3**	47.9	45.1

$$N_t\beta_t + N_p\beta_p + N_b\beta_b = \zeta.$$

- N_t: number of tag-labeled images
- N_p: number of point-labeled images
- N_b: number of box-labeled images
- ζ: fixed total annotation budget

A balanced mixture can outperform either weak label alone under the same annotation cost.

Why Does Heterogeneous Supervision Work?

- Tags Provide Coverage
  - Many images can be labeled.
  - The model observes diverse appearances and contexts.
  - Localization is weak.
- Points and Boxes Provide Localization
  - Fewer images can be annotated.
  - Object locations are explicitly provided.
  - Boxes provide stronger spatial constraints.
- Heterogeneous Labels Provide Both
  - Tags improve data coverage.
  - Points and boxes anchor instance localization.
  - SAM's prompt space transfers information across label types.

Cheap labels provide breadth, while strong labels provide spatial precision.

Does Each Component Help?

- Prompt Matching
  - Aligns object queries with weak annotations in SAM's latent space.
  - Helps the model learn instance identity without mask ground truth.
- Adaptive SAM Mask Matching
  - SAM returns three masks for each prompt.
  - WISH selects the candidate closest to the current prediction.
  - Adaptive selection outperforms using a fixed mask level.
- Self-Enhancement
  - Current instance masks refine the CAM.
  - Improved CAMs generate better point prompts for tag supervision.

Table 4. Ablation study for WISH framework on PASCAL dataset. We verify the impact of $\mathcal{L}_{\text{prompt}}$, $\mathcal{L}_{\text{mask}}$, and $\mathcal{L}_{\text{self}}$ on performance.

Exp	L_{prompt}	L_{mask}	L_{self}	mAP
A		Adaptive	✓	43.6
B	✓	Lv1	✓	36.2
C	✓	Lv2	✓	40.7
D	✓	Lv3	✓	42.8
E	✓	Adaptive		41.5
F (WISH)	✓	Adaptive	✓	**46.0**

Prompt matching, adaptive mask selection, and self-enhancement are complementary.

From One Weak Label to a Portfolio of Labels

- Conventional WSIS
  - Chooses one weak label type for the entire dataset.
  - Treats tags, points, and boxes as separate settings.
- WISH
  - Combines different label types in one model.
  - Uses SAM’s prompt space as a common representation.
  - Matches predictions using class, prompt, and mask information.
  - Achieves better cost–performance trade-offs.

The optimal supervision strategy may be a portfolio of weak labels, not a single annotation type.

Next Week..

- Active learning ..